Certified threshold decisions from error-bounded river water-quality records over one-way satellite IoT links

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Abstract

Satellite Internet of Things services reach remote rivers but sell only a few short, one-way messages per day, and the server never learns which messages were lost. This study asked what a server can soundly decide about water-quality thresholds from the bytes that arrive, and spent the bytes on the decisions a receiver may need rather than on reconstruction. A node sends an error-bounded record within the accuracy of an excellent USGS record, plus witnesses of hard cases: quantised daily extrema and a conductance rate pair. A three-valued rule answers exceeded, not exceeded, or undetermined; the answers are proved correct for any pattern of lost messages under stated conditions. A first confirmation of an earlier version failed its integrity hypothesis and used an unfair comparator; the revised method was confirmed once on 35 sealed rivers (2022). All three pre-registered tests held. On unconfigured thresholds, witnesses decided 0.182 to 0.226 more hard cases than a node sending configured decision flags plus the record, and 0.197 to 0.275 more than the record alone. The cost was 0.026 to 0.082 in whole-record recovery; at six messages per day under independent loss (0.082) it exceeded the design's own adoption limit of 0.08, and the design was adopted post hoc. No certain decision was wrong in the tested configurations. Flags remained better on the thresholds they encode. A 24-bit check guards against random corruption, not deliberate forgery. The evidence is trace-driven simulation with modelled loss.

Keywords: water-quality monitoring, satellite Internet of Things, error-bounded records, certified decisions, erasure coding, pre-registered evaluation

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1. Introduction

Continuous water-quality monitoring underpins the management of rivers that supply drinking water and receive pollution. In-situ sensors measure temperature, specific conductance, dissolved oxygen, and pH every few minutes, and the resulting records reveal diel cycles, storm responses, and short pulses that grab samples miss (Rode et al., 2016). Wireless sensor networks have made such monitoring affordable (Adu-Manu et al., 2017), and deployments in Ghana and Kosovo have shown that it works in practice, but they relied on cellular backhaul (Adu-Manu et al., 2020; Ahmedi and Ahmedi, 2022). The rivers under the greatest pressure are often the furthest from that infrastructure; in Ghana, for example, small-scale gold mining has degraded rivers in forested catchments where terrestrial coverage is sparse (Duncan, 2020). This motivating case is not evaluated here.

Satellite Internet of Things (IoT) services reach such sites with small, low-power terminals (Centenaro et al., 2021), but their capacity is sold in messages, not bandwidth. A typical module sends one-way messages with a payload of 23 bytes and receives no acknowledgement (Kinéis, 2023). A four-parameter sensor sampling every 10 minutes produces 576 readings a day; even packed at 13 bits each, without timestamps, into the 20 bytes that remain after a three-byte message header, they would fill at least 47 messages. Few such messages are affordable, some are lost, and the node is never told which. A monitoring agency therefore needs to know what the server can still conclude from what arrived. Many uses tolerate a delay of a day: compliance assessment, trend analysis, and model calibration work from daily or longer summaries, whereas alarms that require action within minutes are outside the scope of this study.

Agencies do not act on records; they act on decisions such as whether dissolved oxygen fell below 5 mg/L or whether a conductance pulse occurred. Delivering a whole record within an accuracy tolerance (Wagner et al., 2006) serves such decisions only partly. A record within tolerance cannot settle a reading that lies closer to a threshold than the tolerance allows, and when the quota is binding, bytes spent on the whole record are bytes not spent near the threshold. This study therefore took the decision as the object: the server states, for each day and criterion, one of three outcomes (exceeded, not exceeded, or undetermined), and the encoder spends its bytes so that the outcomes that matter are decided, with soundness that holds whatever the loss pattern, provided the messages that arrive are uncorrupted.

The study asked five research questions:

1. RQ1: Under which assumptions is a three-valued threshold decision from re-

ceived bytes sound, and what can no decoder decide?

2. RQ2: Do compact extremum witnesses, and targeted protection of the witnesses of hard cases, raise the share of decided hard cases over a record alone at a fixed byte quota?
3. RQ3: How does the witness design compare with decision flags (the answers to configured thresholds sent as bits), send-on-delta sampling, and a tightened tolerance, including on thresholds that were not known when the node was configured?
4. RQ4: What does it cost, in record recovery and in message capacity?
5. RQ5: How much does a message that is corrupted but passes its check endanger the soundness of the decisions?

Five terms recur throughout:

- Certificate: the server's three-valued answer to a threshold query for one day (exceeded, not exceeded, or undetermined), computed only from the bytes that arrived; a certain answer (exceeded or not exceeded) is correct under the conditions stated in Section 3.
- Witness: a compact value sent beside the record that settles a decision the record alone may not, such as the quantised daily minimum of dissolved oxygen.
- Hard case: a day on which the record alone may not settle a query, because the relevant value lies near the threshold or the threshold is crossed; the bands that define it are given in Section 3.
- Configured and unconfigured thresholds: configured thresholds are known to the node when it is set up; unconfigured thresholds are thresholds the node was not set up for, such as a criterion revised after deployment. In this study the unconfigured thresholds were chosen by the analysts and are close to the configured ones.
- Whole-record recovery: the share of days on which every planned value of the error-bounded record arrived; on such days every reading is certified within tolerance.

The broader principle is that a severely bandwidth-limited monitoring system should keep the information that the decisions of a future receiver need, rather than only the information that reconstructs the record; the study tests one implementation of that principle for four water-quality parameters.

The building blocks are established, and the study does not claim otherwise: minimum-size piecewise-linear approximation within a maximum error (Imai and Iri, 1988; O'Rourke, 1981); budgeted compression of sensor history (Deligiannakis et al., 2007); priority encoding transmission, which protects parts of a message unequally (Albanese et al., 1996; Boucheron and Salamatian, 2000); deter-

ministic interval answers from compressed or uncertain data (Cheng et al., 2003; Brito et al., 2017; Lin et al., 2019a); possible-minimum and possible-maximum filtering of predicates (Taurone et al., 2024); and send-on-delta sampling (Diaz-Cacho et al., 2017). The contributions of this study are the following:

- A certificate language for a one-way lossy link: every piece of received information becomes an interval on a reading; threshold, extremum, conductance-change, and rolling-mean queries receive three-valued answers; the answers are sound when the intervals are valid (proved and checked exhaustively on small cases); and two limits are stated, namely that the rules are exact only without witnesses and that the tolerance makes near-threshold readings undecidable from a record alone.
- A message frame with one CRC-24 per transmitted message, a witness section of 13-bit codes, and a hardened receiver (a sanity rule, corroboration, and rejection of any block whose decoding fails), together with a byte-level codec, fuzz tests, and a mutation check of the tests.
- A comparison of designs on identical loss draws at byte level: record only, targeted witnesses (the 3b rule), and a fair flags comparator that sends the answers to the configured thresholds and also uses the record; in development, an exact allocator, send-on-delta, a tightened tolerance, and two hybrids, including negative results (the exact allocator did not beat the simple rule, and flags win on configured thresholds).
- Two pre-registered confirmations on sealed rivers in years that development had not used. The first (version 2 of the method, 31 rivers, 2024) failed its integrity hypothesis, and an internal review then found that its flags comparator discarded the record; the method was revised (version 3), and the earlier cohort became development data. The second (version 3, 35 rivers, 2022) tested three hypotheses that could fail; all held. A modelling error found before the second freeze, and a design adopted although it missed its own adoption limit, are reported as they occurred.

Section 2 reviews related work. Section 3 describes the method. Section 4 describes the data and the pre-registered design. Section 5 reports the development and confirmation results, Section 6 discusses them and their limits, and Section 7 concludes.

2. Related work

The study combines established ideas, and Table 1 states for each line of work what it shares with this study and where it differs. The comparison is of capabili-

ties and settings, not a claim that earlier work is deficient: most of these systems were designed for interactive networks or storage, where a missing value can be requested again.

2.1. High-frequency water-quality records and their quality

In-situ sensors record river chemistry every few minutes, which reveals storm responses, diel cycles, and short pollution pulses that monthly sampling misses (Rode et al., 2016). Such records are useful only if their quality is known: automated quality assurance and control has become part of streaming environmental data management (Campbell et al., 2013), and open software supports the visualisation, correction, and flagging of continuous records (Horsburgh et al., 2015). Agencies grade records by accuracy; the USGS rates a continuous record as excellent when its readings lie within narrow limits of the reference (Wagner et al., 2006). This study took those limits as the tolerance of the record and did not address sensor quality control, which remains upstream of the encoder.

2.2. Error-bounded reduction of sensor records

Sensor data reduction with guaranteed accuracy has two classical formulations. The first fixes a precision and minimises communication (Olston et al., 2003; Deligiannakis et al., 2004). The second fixes a budget and minimises error: buffered multi-signal history was compressed to a bandwidth budget with a maximum-error variant (Deligiannakis et al., 2007). For a single signal, the fewest vertices of a piecewise-linear approximation within a maximum error can be found efficiently (Imai and Iri, 1988; O'Rourke, 1981), also online (Soroush et al., 2008); swing and slide filters build such approximations online with a proved precision guarantee (Elmeleegy et al., 2009), and bounded-error approximation has been restricted to endpoints on a coarse grid (Lin et al., 2019b, 2024). Recent work preserves statistical features of the series, such as the autocorrelation function, within guaranteed bounds (Muñiz-Cuza et al., 2025), or supports analytics directly on a compressed base (Sun et al., 2025). These methods bound the error of every reading, which is what a threshold decision needs; they assume that what is sent arrives. This study used the minimum-number formulation as the encoder of the record and added witnesses and certificates around it.

2.3. Monitoring under suppression, loss, and uncertainty

The closest problem setting is the monitoring of sensor networks with reduced reporting. In temporal suppression a node reports only when its value leaves a

band, and the sink infers the rest; constraint chaining extends this to spatial suppression (Silberstein et al., 2006), and model-based schemes send a model and report only outliers, with user-controlled error bounds (Tulone and Madden, 2006) or with readings acquired only when a probabilistic model falls below a required confidence (Deshpande et al., 2004). Message loss breaks suppression, because a missing report may be a suppression or a failure; BaySail resolves that ambiguity with Bayesian inference over the suppression scheme and the process model, and returns posterior distributions (Silberstein et al., 2007). This study faces the same ambiguity on a one-way link and resolves it differently: erasure coding and a block identity in every message tell the server which information it holds, and the answers are deterministic and three-valued rather than probabilistic. In-network filtering for type-threshold functions such as the maximum forwards only the readings that can affect the function, trading communication against a bounded error (Riva and Finochietto, 2014); it shares the idea that only some readings settle a decision, in a network where the sink can ask. Growth codes raise the share of sensed data that survives node failures by coding data across neighbouring nodes (Kamra et al., 2006); they maximise data recovered, not decisions certified.

Query processing over uncertain data answers with interval bounds and prunes values that cannot affect the answer (Cheng et al., 2003); sensor databases that store an interval instead of the exact value can return answers that are certain or only possible (Cheng and Prabhakar, 2003). The notion goes back to certain answers over incomplete databases, the answers that hold in every complete database consistent with what is known; a system of operators is called safe when no incorrect conclusion can be derived and complete when every valid conclusion can (Imielinski and Lipski, 1984). The three-valued certificates of this study are certain answers in that sense, over the set of records consistent with the received bytes: Proposition 1 states their safety (soundness), and Proposition 2 their completeness (exactness) without witnesses. Plato and its predecessor evaluate time-series expressions over segment-tree summaries of compressed series and return deterministic error intervals for aggregates (Brito et al., 2017; Lin et al., 2019a). Closest in mechanism, TREAT combines erasure coding with compression and uses the possible minimum and maximum of a value, computed from the bits that were retrieved, to filter predicates before download (Taurone et al., 2024). These works establish that interval answers, certain answers, and possible-extremum filters are classical; they address storage, retrieval, or interactive networks.

2.4. Priority encoding transmission and unequal protection

Priority encoding transmission (PET) lets the sender assign each part of a message a priority, the fraction of the encoding packets that suffices to recover that part. Albanese et al. proved that no system can have a total encoding length below the girth of its priorities and stated that systems reaching the girth exist for a priority function that closely approximates the requested one (Albanese et al., 1996); Boucheron and Salamatian related the approach to the capacity region of the broadcast erasure channel with degraded message sets (Boucheron and Salamatian, 2000). The layered group of witnesses in the frame of this study is an instance of this idea with two priority levels, namely the record, which needs K of Q messages, and a witness group, which needs k_w of Q. This study claims no coding result. PET decides how to protect parts that the sender has already defined; this study decides which parts of a river record to define and protect so that regulatory decisions can be certified.

2.5. Event-triggered telemetry, satellite practice, and task-oriented communication

Send-on-delta sampling transmits a value when it leaves a deadband around the last sent value, with the threshold chosen for a target transmission rate (Diaz-Cacho et al., 2017); it is the standard event-triggered baseline, and this study evaluated it in development and in the first confirmation, alone and inside the record-first erasure code, and, after the second confirmation, on the final frame (Section 5.7). Direct-to-satellite IoT has been studied mainly at the medium-access layer (Capez et al., 2022; Tondo et al., 2021; Alvarez et al., 2022; Afhamisis and Palattella, 2022; Rathi and Borkotoky, 2025), with packets treated as interchangeable. The closest operational practice comes from animal telemetry: satellite relay data loggers abstract each ocean profile to a fixed number of broken-stick points and send the messages pseudo-randomly through the one-way Argos system (Boehme et al., 2009; Photopoulou et al., 2015). That practice fixes the number of points rather than the accuracy or the decision. Goal-oriented and semantic communication transmit what a downstream task needs rather than the data themselves (Zhang et al., 2023), and age-of-information research optimises the timeliness or the task performance of the latest value (Yates et al., 2021; Chen et al., 2021; Talli et al., 2026); this study shares that objective, for past threshold decisions, with deterministic guarantees instead of learned or average-case ones.

2.6. Reproducible environmental software

Environmental modelling has long asked that models and software be developed and evaluated transparently (Jakeman et al., 2006), and computational hydrology has been criticised for results that cannot be reproduced (Hutton et al., 2016). Recent work in this journal defined levels of reproducibility (Essawy et al., 2020; Choi et al., 2021). This study therefore separated development from confirmation, froze its software and hypotheses before the confirmation data were opened (in the second confirmation, before they were downloaded), and reports both of its confirmations, including the hypothesis that failed in the first and the changes made between them.

2.7. Position of this study

The contribution is a source-side, byte-budgeted integration of tolerance-bounded time-series reconstruction, targeted extremum witnesses, and unequal erasure protection, with a receiver that returns conditional three-valued certificates for environmental threshold queries over a one-way link. It is a systems contribution: no new coding result and no new theory of threshold queries is claimed.

3. Method

Fig. 1 summarises the method. A sensor node at a remote river site records four parameters. At the end of each day it builds a record of the day, adds witnesses, packs both into messages with erasure-coded redundancy, and sends a fixed number Q of messages through a one-way satellite link. The link delivers each message or loses it, and the node never learns which. The server decodes what arrived and answers, for each criterion, exceeded, not exceeded, or undetermined. This section defines the record (Section 3.1), the certificate language and what it can and cannot decide (Section 3.2), the witnesses (Section 3.3), the message frame and receiver (Section 3.4), and the designs compared (Section 3.5).

3.1. Record, tolerance, and encoder

A channel-day is one parameter of one site on one UTC day, with readings $x(t)$, $t = 0, \ldots, n-1$, on the parameter's own sampling grid. The tolerance of a reading is

$$\delta(x) = \max(a,\ r|x|), \tag{1}$$

Table 1: Related work: what each line of work shares with this study and where it differs.

Line of work	Shared with this study	Different setting or aim
Error-bounded piecewise-linear approximation (Imai and Iri, 1988; O'Rourke, 1981; Elmeleegy et al., 2009; Lin et al., 2024)	A maximum error on every reading, the basis of the record	Assumes delivery; no decisions or quota
Budget-first history compression (Deligiannakis et al., 2007)	A bandwidth budget with a maximum-error variant	Interactive network; no certificates
Suppression and model-based monitoring (Silberstein et al., 2006, 2007; Tulone and Madden, 2006; Deshpande et al., 2004)	Reduced reporting; ambiguity between suppression and loss	Probabilistic inference or acknowledged links; this study uses coding and deterministic answers
Filtering for threshold functions (Riva and Finochietto, 2014)	Only readings that can affect a threshold function are sent	In-network, query-driven, bounded error rather than certificates
Data persistence under failures (Kamra et al., 2006)	Coding so that data survive losses	Maximises data recovered, not decisions
Interval and certain answers (Imielinski and Lipski, 1984; Cheng et al., 2003; Cheng and Prabhakar, 2003; Brito et al., 2017; Lin et al., 2019a)	Answers that hold for every consistent state	Storage or querying; no choice of what to send
Erasure coding with predicate filters (Taurone et al., 2024)	Possible-minimum and possible-maximum filters over coded data	Retrieval with requests; no one-way quota
Priority encoding transmission (Albanese et al., 1996; Boucheron and Salamatian, 2000)	Unequal protection with recovery from any fraction of packets	Parts given by the sender; no decision model
Send-on-delta and satellite tag practice (Diaz-Cacho et al., 2017; Boehme et al., 2009; Photopoulou et al., 2015)	Event-triggered or summarised telemetry over thin links	Fixed point counts or deadbands; no certificates
Semantic and goal-oriented communication (Zhang et al., 2023; Yates et al., 2021)	Transmit what the task needs	Learned or average-case objectives

where the absolute part a and the relative part r follow the accuracy that the USGS requires of an excellent continuous record (Wagner et al., 2006): 0.2 °C for temperature; 3% of the value for specific conductance; the greater of 0.3 mg/L and 5% of the value for dissolved oxygen; and 0.2 units for pH. The tolerance is relative to the sensor record; the accuracy of the sensor itself (calibration, fouling, and drift) is outside it.

The encoder selects the fewest kept slots whose piecewise-linear reconstruction keeps every reading within tolerance. This is the minimum-number problem of polygonal approximation (Imai and Iri, 1988) with per-reading tolerances,

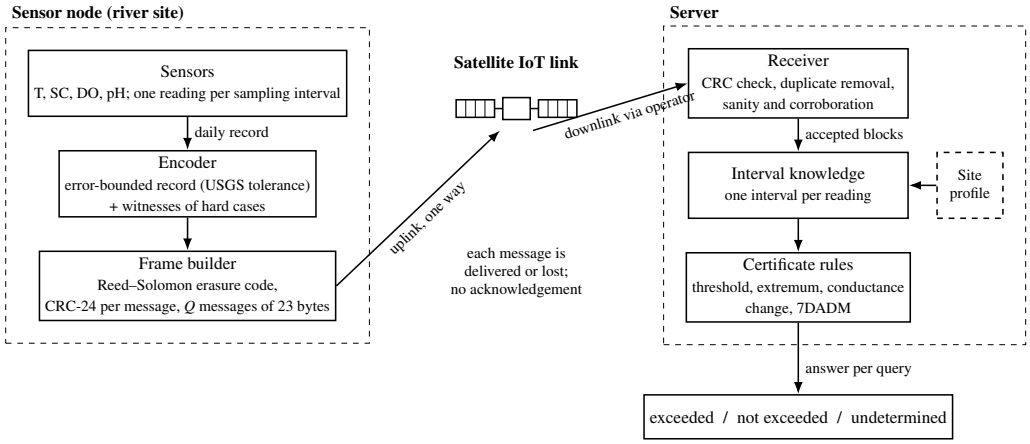


Figure 1: Architecture of the method. The sensor node encodes each day into Q messages; the one-way satellite link delivers or loses each message without acknowledgement; the server decides from accepted messages and the site profile only.

solved exactly by a forward dynamic program over slope windows (O'Rourke, 1981) in $O(n^2)$ time and $O(n)$ memory per channel-day. Three features make the bound exact and the record self-contained:

- Quantisation: every candidate error is computed with the dequantised endpoint values that the server will hold, such that the bound includes quantisation and needs no margin.
- Head and tail: the first kept value is held back to midnight and the last is held to the end of the day, such that no value of the previous day is needed.
- Censoring: a reading outside the quantiser range is always kept, carried as a flag code, excluded from every bound and decision, and reported as below or above range.

A kept reading is a 13-bit code. Linear grids are decimal (0.01 °C, 0.005 mg/L, and 0.002 pH units), and specific conductance uses a logarithmic grid with a step of 0.14% of the value; the worst quantisation error is 2.5% of the tolerance.

3.2. Certificate language

The language is independent of the codec: every source of information becomes a closed interval on a reading, and decisions are computed from intervals alone. The server holds, for every reading, an interval $[\mathrm{lo}(t), \mathrm{hi}(t)]$, a set of readings known to be censored, and optional global bounds. The knowledge is valid when every uncensored reading lies in its interval and every global bound holds.

Truth statements refer to the sensor record that the loader produced, not to the river.

Sources of information and the intervals they provide:

- Base range: the quantiser range of the parameter, valid by the definition of censoring.
- Kept value with code c: the quantiser cell of c, which is far narrower than the tolerance (0 to 0.0025 units for temperature, dissolved oxygen, and pH, and 0.07% for conductance).
- Certified span: the tolerance interval of Eq. (1) around the reconstruction, valid when the day's bound flag is true and the bounding values of the span arrived. Lemma 1: for $r < 1$ and $x \geq 0$, the set $\{x : |x - \hat{x}| \leq \max(a, rx)\}$ is the interval

$$\left[\min\left(\hat{x} - a,\ \frac{\hat{x}}{1+r}\right),\ \max\left(\hat{x} + a,\ \frac{\hat{x}}{1-r}\right)\right]. \tag{2}$$

  The two sets whose union defines it share the point $\hat{x}$, such that the union is an interval with the extreme endpoints.
- Minimum witness: the cell of the kept extremum value, together with the assertion that every uncensored reading of the day is at least the lower end of that cell; the maximum witness is symmetric.

A reading with no information has the base range, such that it blocks a certain no.

Queries have three-valued answers: 1 (certainly yes), 0 (certainly no), or -1 (undetermined). With global bounds g_{lo} and g_{hi} (from witnesses), the effective interval of reading t is

$$\mathrm{lo}_{\mathrm{eff}}(t) = \max(\mathrm{lo}(t),\ g_{lo}), \qquad \mathrm{hi}_{\mathrm{eff}}(t) = \min(\mathrm{hi}(t),\ g_{hi}). \tag{3}$$

The rules are as follows:

- Exceeds below θ (low dissolved oxygen, low pH):

$$A_\theta = \begin{cases} 1 & \text{if } \mathrm{hi}_{\mathrm{eff}}(t) < \theta \text{ for some } t, \\ 0 & \text{if } \mathrm{lo}_{\mathrm{eff}}(t) \geq \theta \text{ for every } t, \\ -1 & \text{otherwise.} \end{cases} \tag{4}$$

  The rule for exceeding above θ is symmetric.
- Day minimum and maximum:

$$\begin{aligned} \min_t x(t) &\in \left[\min_t \mathrm{lo}_{\mathrm{eff}}(t),\ \min_t \mathrm{hi}_{\mathrm{eff}}(t)\right], \\ \max_t x(t) &\in \left[\max_t \mathrm{lo}_{\mathrm{eff}}(t),\ \max_t \mathrm{hi}_{\mathrm{eff}}(t)\right]. \end{aligned} \tag{5}$$

- Conductance change: some pair of readings one hour apart with a relative change above 10%. For readings a and b with intervals $[L_a, U_a]$ and $[L_b, U_b]$,

$$\frac{x(b)}{x(a)} \in \left[\frac{L_b}{U_a},\ \frac{U_b}{L_a}\right], \tag{6}$$

  and both ends are attained. A certain no holds when every pair is within 10% for all values of the intervals. A certain yes holds when no assignment of readings within their intervals keeps every pair within 10%; for one lag, the pairs form chains, and the set of readings reachable under "all pairs within 10%" is propagated along each chain until it is empty. The pairwise rule "some single pair is certainly changed" is sound but incomplete (readings 100, an interval of 100 to 130, and 130 always contain a change, but no pair is certain); therefore, the chain rule is used.
- Seven-day mean of daily maxima above 20 °C (7DADM): with daily-maximum intervals $[m_d^{lo}, m_d^{hi}]$ for the seven days d of the window,

$$\frac{1}{7}\sum_d \max_t x_d(t) \in \left[\frac{1}{7}\sum_d m_d^{lo},\ \frac{1}{7}\sum_d m_d^{hi}\right]. \tag{7}$$

  A day with no information contributes the base range, never zero and never omission; one missing day therefore blocks a certain no but not a certain yes that the other six days already force.

Let V be the uncensored readings of a channel-day (the domain of every query) and $S \supseteq V$ the readings that the server treats as uncensored; a reading in $S \setminus V$ is censored but not known to the server, because it lies inside a certified span and its own code did not arrive. The knowledge is valid on V when every reading in V lies in its interval, every global bound holds on V, and every reading that the server holds as censored is censored.

Proposition 1 (soundness of the rules of Eqs. (3) to (7)): let the knowledge be valid on V.

1. Every answer 0 of the threshold rules and of the conductance rule, the lower end of every day-minimum interval, and the upper end of every day-maximum interval are correct, whatever $S \setminus V$ is.
2. Condition B: every reading t in $S \setminus V$ has an interval that reaches the values of genuine readings on both sides, that is, $\mathrm{lo}(t) \le x(u)$ and $\mathrm{hi}(t) \ge x(v)$ for some u, v in V. Under condition B, every answer 1 of the threshold rules, both ends of the day-extremum intervals, and every answer of the 7DADM rule are also correct.

3. Every answer 1 of the conductance rule is correct when no conductance reading of the day is in $S \setminus V$.

Lemma 2 (when condition B holds): a reading in $S \setminus V$ whose certified span is bounded by kept values with uncensored codes satisfies condition B, because its reconstruction lies between the dequantised bounding values, each within half a quantiser step of its true reading, and the half-width of the tolerance interval exceeds half a step for every parameter. Condition B can fail only when a bounding kept value is itself censored; the reconstruction is then pulled to that edge of the quantiser range, and only the queries on that side (below and minimum at the low edge, above and maximum at the high edge) are affected. The proofs are in Appendix A. As a check, an exhaustive test over every subset of arrived values on small days, with and without each witness, found no interval that excluded the truth and no wrong answer.

The empirical scope of the conditions was counted after the confirmation (post hoc). In the three cohorts (30,854 analysed days), censored readings occurred on 4 days, all at one 2022 site (05576100) and all small negative dissolved-oxygen values; no conductance reading was censored; therefore, part 3 was never at risk. On those 4 days condition B could fail on the low side; the true answer to every dissolved-oxygen query on them was yes at all three threshold sets, and a low-side failure can only add a spurious yes. No reported decision therefore depended on a case that Proposition 1 does not cover.

Proposition 2 (exactness without witnesses, and its failure with them): for the threshold queries and the day extrema, when no witness has arrived, the rule returns an answer other than -1 exactly when the query has the same value on all records consistent with the knowledge. With a witness, the rules are sound but not exact. Counterexample: reading 1 in $[3, 10]$, reading 2 in $[4.5, 4.6]$, and a minimum-witness cell $[4.0, 4.01]$. Reading 2 cannot be the minimum; therefore, reading 1 lies in $[4.0, 4.01]$ and "some reading above 5" is certainly false, but the box rule returns -1. The decisions reported in this study are therefore a lower bound on what the witnesses could decide.

Proposition 3 (identifiability limit): if two records consistent with the received bytes have different answers to a query, no sound decoder can decide that query from those bytes. The tolerance makes this concrete. From a certified span alone, a threshold θ is decidable at a reading only when the reconstruction is farther from θ than the half-width of the interval (0.2 pH units, 0.2 °C, 3% of conductance, and the greater of 0.3 mg/L and 5% for dissolved oxygen). A reading closer to the threshold can be decided only if its value is carried, because the cell of a kept value is orders of magnitude narrower. This is the formal reason why a record within

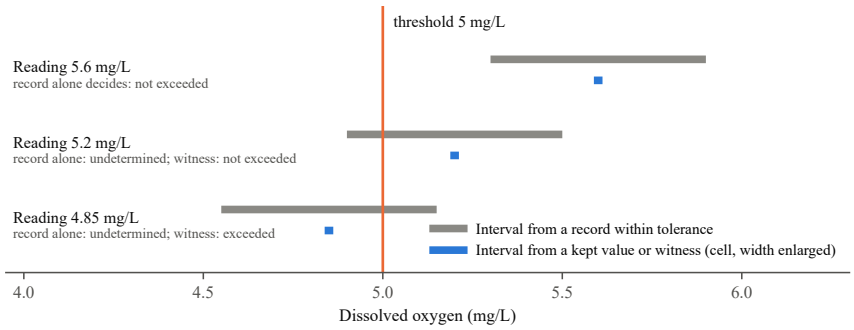


Figure 2: Constructed illustration (not data) of the identifiability limit for dissolved oxygen against a threshold of 5 mg/L. The tolerance interval of a record (grey) decides a reading far from the threshold but leaves readings near it undetermined; the cell of a kept value or witness (blue, enlarged for visibility) decides them.

tolerance misses near-threshold events, and why bytes belong near the threshold and not everywhere. Fig. 2 illustrates the point with constructed values.

Soundness is conditional, not unconditional. It requires correct and corroborated messages, an honest encoder, a correct site profile, the quantiser range, and, where the rules use it, assumption R: readings are multiples of the site's reporting resolution. In this study the resolution was inferred from the analysed record itself, and the record contains readings that the loader bridged linearly over gaps of at most two readings. A wrong resolution is unsound: a reading of 7.03 with an assumed resolution of 0.1 yields a certain wrong decision for the threshold 7.01. In development, no resolution could be inferred for dissolved oxygen and pH at 15 of 38 sites, where assumption R was therefore off; in the second confirmation a resolution was inferred for at least one of dissolved oxygen, pH, and temperature at 26 of 35 sites (dissolved oxygen 14, pH 23, temperature 11).

3.3. Witnesses and hard cases

A witness is the quantised extremum of one day and one parameter: the minimum of dissolved oxygen, the minimum and maximum of pH, the maximum of temperature (for the 7DADM), and the conductance rate pair, the two kept values that realise the day's largest hourly relative change. A witness costs 13 bits per code. It carries no slot, because no decision rule uses one.

Not every day needs its witnesses. A case is hard when the record alone is likely to leave the decision open:

Table 2: Message frame (23-byte payload).

Part	Bits	Content
Header (not coded)	24	version (2); block day modulo 64 (6); message index (7); K, the record data units (7); coded flag (1); witness-group flag (1)
Check	24	CRC-24 over the header and the rest of the body, in every message
Witness descriptor (if flagged)	8	k_w, the data units of the witness group (3); presence mask of the five witness objects (5)
Witness symbol (if flagged)	$8s_w$	one symbol of the witness group, with $s_w = \lceil \lceil 13 \cdot \text{codes}/k_w \rceil / 8 \rceil$ bytes
Record symbol	rest	one Reed–Solomon symbol of the record data units

- within 0.6 mg/L of the dissolved-oxygen threshold, within 0.4 pH units of a pH threshold, or in the event state;
- a conductance rate of at least 4% per hour;
- a 7DADM trailing mean within 3 °C of 20 °C, or with unknown history.

These bands were fixed on development data (the 7DADM band and the unknown-history rule after a first run fell short), and the confirmation used them unchanged.

3.4. Message frame and receiver

Each message carries the 23-byte user payload of the KIM1 module (Kinéis, 2023); the operator adds the device identifier and the time of reception (Kinéis, 2024). Table 2 provides the layout of the frame. A three-byte header (version, block day modulo 64, message index, K, and two flags) is not erasure-coded. The 20-byte body carries a CRC-24 (polynomial 0x864CFB) over the header and the rest of the body of every transmitted message, whether data or parity, an optional witness section, and one Reed–Solomon symbol of the record units over GF(2^8). The block day is unwrapped from the operator's reception time.

A record data unit carries up to three segments of kept values and a hosted-witness section: a flag, a 5-bit presence mask, and 13-bit codes in the spare room of the unit. Hosted witnesses cost no record capacity. Without witnesses the 136-bit record symbol holds four record values of 24 bits after 24 bits of unit overhead, which leaves 16 spare bits; a layered witness group takes whole bytes from the record symbol.

The check length was chosen on development data by a rule written before the comparison: four candidate frames (CRC-16 as in the earlier version, CRC-16 with a record-level CRC-32, CRC-24, and both) were compared by the expected number of days with a false certificate per million days at an assumed bit-error rate of 10^{-3}, with a limit of 0.5, and among the frames that met the limit the one with the highest whole-record recovery, with ties going to the simpler frame, was selected. Only the two CRC-24 frames met the limit, and the simpler one was selected. The CRC-24 costs one record value per message (four instead of five) and lowers the probability that a corrupted message passes the check by a factor of 256.

The receiver applies these rules:

1. Accept a message only if its CRC passes, and drop duplicates; an object that is not a message of the right length counts as a failed message.
2. Decode the record from any K accepted messages and the witness group from any k_w accepted messages; with a check on every message there is no retry over combinations of parity messages.
3. Sanity: reject values outside the site profile or out of planned order, and censored witness codes.
4. Corroboration: when more messages arrive than are needed, re-encode the decoded units and compare with every other accepted message; any mismatch rejects the block, and nothing is certified from it. A witness descriptor that differs between messages, or two CRC-valid copies of one message that disagree, also reject the block.
5. Any failure while decoding a block, of whatever kind, rejects the record and the witnesses of that block.
6. Produce certificates by the rules above from accepted, corroborated content only. A day on which no reading is uncensored has no defined extremum, and its extremum interval is unbounded, which leaves every decision that needs it undetermined.

A limit remains: a record value with a censored code (the lowest or highest code) passes the sanity rule, because honest records carry such codes for readings outside the quantiser range, and it removes that reading from every decision. Section 5.6 reports how often a forged message led to a wrong decision.

3.5. Designs compared

The confirmation compared three designs on the same loss draws at byte level: real messages were built, lost, decoded by the receiver, and the decisions were made from the decoded values and witness codes only.

- Record only: the record, with D_1 data units and $Q - D_1$ parity messages of a systematic Reed–Solomon code, such that any D_1 messages recover it.
- 3b rule: the generalised rule of this study. Witnesses of hard cases form a layered group in their own protected message columns ($k_w \leq 2$); witnesses of easy cases are hosted in the spare room of the last record unit. In the language of priority encoding transmission (Albanese et al., 1996), the record and the witness group are two priority levels over the same Q messages.
- Flags plus record: the record-only blocks plus the answers to the four configured queries (4 bits) in every message. The flag field fits in the 16 spare bits of the record symbol; therefore, the record is the same as in the record-only design. The server takes the certificate decision when it is certain and otherwise the configured answer whenever any message arrived; an unconfigured threshold is decided by the record or by monotone implication from the configured answer. The flag answers are taken from the true record; therefore, this comparator is idealised; it is the fair comparator for the witness design because it uses everything a flag node would receive.

In development, the following were also compared in an abstract slot model (a message holds a whole number of value slots, a group is a systematic maximum-distance-separable code, and witnesses occupy whole slots): decision flags that do not use the record, an exact allocator of witnesses to columns, two hybrids of flags and witnesses, the same layering on random days (a control for targeting), send-on-delta with hold reconstruction (Diaz-Cacho et al., 2017), and the record at half the tolerance. These were not re-run in the final version of the method, and no claim rests on them (Section 5.1). The slot model was also run in the confirmation as a cross-check of the byte-level results; it assigns the witness group whole slots and therefore overstates the record cost of the 3b rule.

4. Data and pre-registered design

4.1. Data and cohorts

All data were approved instantaneous values from the USGS National Water Information System (U.S. Geological Survey, 2016); dissolved oxygen was used in mg/L. Each parameter was used on its own sampling grid on each UTC day; a run of at most two missing readings with valid neighbours on both sides was bridged linearly, and any longer gap made that parameter-day ineligible. The analysis used four-parameter days. The study passed through two rounds of development and confirmation:

- Development, first round: calendar year 2023 at 38 sites with four-parameter days in New Jersey, Pennsylvania, Illinois, and Oregon. The bands, rules, and hypotheses of version 2 of the method were set on these data.
- First confirmation (version 2): calendar year 2024 at 31 sealed sites in four regions, selected from metadata only. Its integrity hypothesis failed and an internal review found that its flags comparator discarded the recovered record (Section 5.2). After that run the cohort became development data for version 3; it was used for the second rehearsal of the final analysis and for no design choice that the second confirmation tested.
- Second confirmation (version 3): calendar year 2022 at 35 sites that no earlier round had used (106 sites excluded). Selection used metadata only: sites whose series of temperature, conductance, dissolved oxygen, and pH covered 2022 and that had at least 250 approved daily means per parameter (qualifiers read, values discarded) were taken in seeded random order within nine regions, at most five per region. Six regions supplied five sites; the Gulf and lower Mississippi and Texas supplied two each and the Northeast and New England one, because few candidates met the approval rule, and the shortfall was not repaired. The data were downloaded only after the freeze, cleaned by the rules of all earlier cohorts, and sealed with MD5 checksums before any analysis. All 35 sites had at least 100 four-parameter days and formed the analysis population (10,417 days, 100 to 361 per site; listed in Table B.1, Appendix B).

4.2. Loss models, plans, and thresholds

No measured loss rate was available for the target service; therefore, loss was modelled. Independent loss removed each message with probability 0.3; a two-state Gilbert channel (Gilbert, 1960; Elliott, 1963) lost every message in its bad state with the same mean loss 0.3 and mean bursts of three messages. Plans of $Q = 6$ and $Q = 12$ messages per day were evaluated, with 20 loss draws per day and condition, and the same draws were used for every design on that day (paired).

The configured thresholds were fixed before any design was compared: dissolved oxygen below 5 mg/L (U.S. Environmental Protection Agency, 1986a); pH below 6.5 or above 9.0 (U.S. Environmental Protection Agency, 1986b); a 7DADM above 20 °C (U.S. Environmental Protection Agency, 2003); and a conductance change above 10% in one hour (a criterion defined for this study). These values are examples of commonly used criteria, not universal standards: water-quality criteria depend on the designated use of a water body (for example cold-water fisheries for the 7DADM), on the region, and on the season, and agencies set their own. The conductance criterion is an event detector, not a regulatory limit:

a rapid change in conductance marks a pulse of road salt, a discharge, or a storm, and 10% in one hour was chosen as a change well beyond the 3% measurement tolerance of conductance. Two alternative threshold sets that the encoder does not know measure generality: alt1 (4.5 mg/L; pH 6.3 and 9.2; 21 °C; 20%) and alt2 (5.5 mg/L; pH 6.7 and 8.8; 19 °C; 5%). The analyst chose these sets in the first round, and they were seen on every development cohort; the results that depend on them are reported with that caveat.

4.3. Hypotheses, freeze, and procedure

The hypotheses of the second confirmation (Table 3) were drafted on 2023 data, rehearsed with the final analysis code on 2023 and on the first confirmation cohort, and reviewed internally. That review found a modelling error in the comparator (Section 5.2); the code was corrected, the hypotheses that depended on the error were dropped or relabelled, and both rehearsals were repeated. The limits of the three tests were written before the rehearsals and were not changed after them; a proposal to widen two limits after the first rehearsal was withdrawn.

Only three hypotheses test the method (H2 to H4). H1 and H8 are checks of the code: H1 cannot fail through assumption R, because the reporting resolution is inferred from the analysed record, and H8 compares two implementations of the same code. H6 is a known limit that holds largely by construction, because the flags comparator takes its configured answers from the truth whenever a message arrives. H7 is an analytic bound: even if every forged message that passed the check produced a false certificate, the expected rate would be 0.042 (at $Q = 6$) and 0.084 (at $Q = 12$) days per million days at the assumed bit-error rate; therefore, no limit on it could fail.

The plan, the analysis code, and the download script were then frozen: the repository was committed and tagged, a freeze record with checksums of 124 files and the test result was written and committed, and only then were the cohort data downloaded and sealed. The analysis script refused to start unless the code equalled the tag, the freeze record was committed and named the tagged commit, the limits in the code equalled a machine-read line of the plan, the seal covered exactly the cohort folder and verified, and a run stamp showed that no earlier output came from other code. The sealed data were then analysed once. The run was interrupted once, when the computer restarted during the byte-level stage; it was resumed with the same command and the same frozen code, which passed every start check again and reused the outputs of finished sites.

Intervals are 95% percentile intervals from 1,000 site-clustered bootstrap resamples; differences between designs are pooled day-query means computed on

the same resamples. The estimand of H2 and H3 is the share of all hard day-queries in the cohort (dissolved oxygen, pH, conductance, and 7DADM windows together, each loss draw counted) that a design decides correctly; it weights each site by its number of hard cases and each query kind by its frequency. Section 5.4 reports, post hoc, the same contrasts with equal weight per site and per query kind. Hard-case decisions pool dissolved oxygen, pH, conductance, and 7DADM windows, and a decision counts as correct only when it is certain and equals the truth on the sensor record. A hypothesis counts as supported only if its stated bound holds in every listed condition.

Table 3: Pre-registered hypotheses of the second confirmation and their outcome on the 35 sealed sites (byte level; ind. is independent loss and Gil. is Gilbert loss).

ID	Type	Statement	Limit	Result (95% interval)	Outcome
H1	Code check	No wrong certain decision; no decoded value differing from the plan; no decision-layer error; no round-trip mismatch	all 0	0 wrong decisions; 0 mismatches in 41,668 blocks	Pass
H2	Test	3b rule minus flags plus record, hard cases at unconfigured thresholds	lower bound at least 0.10	$Q = 6$, ind.: +0.226 (+0.196 to +0.258); $Q = 6$, Gil.: +0.189 (+0.165 to +0.215); $Q = 12$, ind.: +0.182 (+0.154 to +0.214); $Q = 12$, Gil.: +0.191 (+0.164 to +0.222)	Pass
H3	Test	3b rule minus record only, hard cases at unconfigured thresholds	lower bound at least 0.10	$Q = 6$, ind.: +0.275 (+0.233 to +0.324); $Q = 6$, Gil.: +0.249 (+0.212 to +0.292); $Q = 12$, ind.: +0.197 (+0.160 to +0.246); $Q = 12$, Gil.: +0.217 (+0.181 to +0.263)	Pass

ID	Type	Statement	Limit	Result (95% interval)	Outcome
H4	Test	Whole record recovered, 3b rule minus record only (the cost of the adopted design)	lower bound at least -0.15 ($Q = 6$), -0.10 ($Q = 12$)	Q = 6, ind.: -0.082 (-0.099 to -0.064); Q = 6, Gil.: -0.068 (-0.081 to -0.055); Q = 12, ind.: -0.026 (-0.035 to -0.017); Q = 12, Gil.: -0.032 (-0.041 to -0.023)	Pass
H6	Known limit	3b rule minus flags plus record, hard cases at configured thresholds, $Q = 6$	upper bound below 0	independent: -0.028 (-0.034 to -0.022); Gilbert: -0.107 (-0.117 to -0.099)	Holds
H7	Analytic bound	Expected false-certificate days per million days, bit-error rate 10^{-3}	none (cannot fail)	3b rule, $Q = 6$: 0.0030 (upper 0.0035); 3b rule, $Q = 12$: 0.0008 (upper 0.0012); record, $Q = 6$: 0.0022 (upper 0.0026); record, $Q = 12$: 0.0006 (upper 0.0009)	Reported
H8	Code check	Record only: byte frame and slot model agree in whole-record recovery	absolute difference at most 0.01	largest absolute difference 0.001	Pass

H5 of the draft (the 3b rule minus flags plus record in whole-record recovery) was dropped before the freeze because, with the flags modelled correctly, it equals H4.

5. Results

Section 5.1 summarises development, Section 5.2 the first confirmation and why the method changed, and Sections 5.3 to 5.6 the single run of the second confirmation on the 35 sealed sites. Values are given at $Q = 6$ under independent loss with loss probability 0.3 unless stated, and all confirmation values are at byte level.

5.1. Development results

On the 2023 development data, the findings that shaped the method were the following (slot model unless stated):

- Soundness: no certificate design gave a wrong decision in any condition.
- Witnesses added decisions: a witness placed inside the record code did almost nothing for the 7DADM, which needs seven informative days; placement in a protected group mattered.
- Targeting paid: against the same layering on random days, the targeted rule gained 0.10 to 0.15 on near 7DADM windows and 0.17 on hard conductance days.
- The exact allocator did not beat the simple rule on hard cases; the simple rule was within 7% (at $Q = 6$) and 1.6% (at $Q = 12$) of the optimum of the stated objective.
- Send-on-delta needed 1.79 times as many values as the piecewise-linear record, and a tightened tolerance did not improve hard-case decisions at this quota.
- With the flags allowed to use the record, hosting all witnesses in the spare room of the flag design (hybrid 1) added only 0.02 to 0.03 on unconfigured thresholds, and adding the 3b rule to the flags (hybrid 2) lost 0.16 to 0.19 of record recovery against the record alone at $Q = 6$. Of the layered designs, only the 3b rule met the adoption rule written before the run (a gain of at least 0.10 over the fair comparator in all four conditions, and a record cost of at most 0.08 against the record alone).
- Moving from CRC-16 to CRC-24 cost one record value per message. On the CRC-24 frame the 3b rule's record cost at $Q = 6$ was 0.088 (independent loss) at byte level, beyond the 0.08 of the adoption rule. The 3b rule was adopted nevertheless. An earlier justification, that the cost was small against the fair comparator, rested on the modelling error described in Section 5.2 and was withdrawn; the adoption is therefore post hoc, and its cost became hypothesis H4.

5.2. The first confirmation and the revision of the method

Version 2 of the method (CRC-16 frame, slot-model comparisons, nine hypotheses) was confirmed once on 31 sealed sites of 2024. Eight hypotheses held, including no wrong decision in 30,285 byte-level round trips and a gain of the 3b rule over the record alone. The integrity hypothesis failed: after one forged message that passed the CRC-16, the upper bounds of the share of days with a false certificate exceeded their limits at $Q = 12$ (0.0134 against 0.01 for the record alone and 0.0215 against 0.02 for layered witnesses). An internal review

then found that the flags comparator of that study discarded the recovered record, which made the contrast against flags unfair, that several hypotheses held by construction, and that the receiver caught only some error types. These findings led to version 3: a CRC-24 per message, a hardened receiver, a flags comparator that uses the record, comparisons at byte level, and a new sealed cohort. The full results of the first confirmation are in Appendix C (Table C.1).

A second internal reading, of the version 3 plan before its freeze, found that the slot model had charged the flags one value slot per message, as on the CRC-16 frame, although the CRC-24 record symbol has 16 spare bits that hold the four flag bits. With the flags modelled correctly, the flags-plus-record design recovers the record exactly as often as the record alone. The draft's hypothesis on the record cost against that comparator was therefore dropped, the comparisons were moved to byte level with paired loss draws, and both rehearsals were repeated before the freeze. The error had no effect on the benefit hypotheses, which stayed above their limits on both development years (Table 4).

5.3. Second confirmation: the three tests held

Table 3 lists every hypothesis with its type, limit, result, and outcome. All three tests held, the two code checks passed, and the known limit replicated (Fig. 3).

- H2, benefit over the fair comparator: on thresholds that the encoder did not know, the 3b rule decided 0.226 (0.196 to 0.258) more hard cases than flags plus record at $Q = 6$ under independent loss and 0.189 (0.165 to 0.215) under Gilbert loss; at $Q = 12$, 0.182 (0.154 to 0.214) and 0.191 (0.164 to 0.222). The smallest lower bound was 0.054 above the limit of 0.10.
- H3, benefit over the record alone: 0.275 (0.233 to 0.324), 0.249 (0.212 to 0.292), 0.197 (0.160 to 0.246), and 0.217 (0.181 to 0.263).
- H4, the cost of the adopted design: the 3b rule recovered the whole record less often than the record alone, by 0.082 (0.064 to 0.099) and 0.068 (0.055 to 0.081) at $Q = 6$, and by 0.026 (0.017 to 0.035) and 0.032 (0.023 to 0.041) at $Q = 12$. The lower bounds were within the limits of 0.15 and 0.10. The point cost at $Q = 6$ under independent loss, 0.082, was again beyond the 0.08 of the original adoption rule; H4 shows that the cost is bounded, not that the adoption rule was met.
- H1, soundness: no certain decision was wrong in any design, plan, or loss model, no decoded value differed from the plan, no decision-layer error occurred, and 41,668 blocks (10,417 days, two designs, two plans, all messages delivered) passed the byte-level round trip. The caveat of Section 4.3 applies.

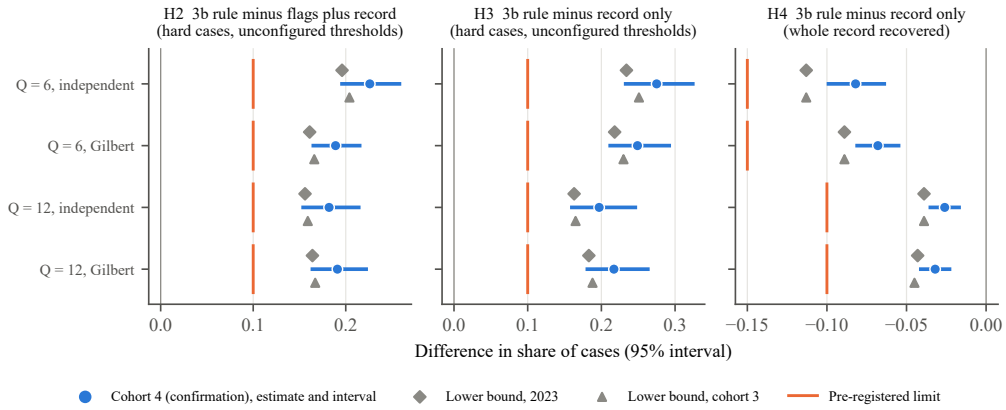


Figure 3: The three tests on the sealed cohort with 95% site-clustered intervals, the lower bounds of the same tests on the two development years (grey), and the pre-registered limits (orange).

- H6, the known limit: at the configured thresholds the flags-plus-record design decided more hard cases than the 3b rule, by 0.028 (0.022 to 0.034) under independent loss and 0.107 (0.099 to 0.117) under Gilbert loss.
- H8, code check: for the record alone, the byte frame and the slot model agreed in whole-record recovery within 0.001.

The confirmation cohort fell within the range of the two development years on every test (Table 4). The cost was slightly smaller than in development (0.082 against 0.088 and 0.094 at $Q = 6$ under independent loss).

5.4. Robustness of the estimates (post hoc)

The confirmatory estimates were re-analysed after the run from the run's own per-site outputs; nothing was re-simulated. The conclusions did not depend on the weighting or the number of resamples:

- With 10,000 resamples instead of 1,000, every interval of H2 to H4 changed by at most 0.002 at either end.
- With equal weight per site, the H2 contrasts were 0.178 to 0.217 (lowest lower bound 0.154) and the H3 contrasts 0.191 to 0.258 (lowest lower bound 0.159); the H4 costs were 0.025 to 0.080.
- With equal weight per query kind, the H2 contrasts were 0.148 to 0.209 (lowest lower bound 0.125) and the H3 contrasts 0.164 to 0.262 (lowest lower bound 0.130).
- Every one of the 35 sites had a positive H2 and H3 contrast in all four conditions (Fig. 4); for H2 at $Q = 6$ under independent loss the 5th, 50th, and 95th

Table 4: The three tests on the two development years and on the confirmation cohort, computed by the same frozen code (point estimate with the lower bound of the 95% interval in brackets; ind. is independent loss and Gil. is Gilbert loss).

Test	Year	$Q = 6$, ind.	$Q = 6$, Gil.	$Q = 12$, ind.	$Q = 12$, Gil.
H2, 3b minus flags plus record	2023 (development)	+0.223 (+0.196)	+0.185 (+0.161)	+0.179 (+0.156)	+0.188 (+0.164)
	2024, cohort 3 (development)	+0.246 (+0.204)	+0.200 (+0.166)	+0.203 (+0.159)	+0.209 (+0.167)
	2022, cohort 4 (confirmation)	+0.226 (+0.196)	+0.189 (+0.165)	+0.182 (+0.154)	+0.191 (+0.164)
H3, 3b minus record only	2023 (development)	+0.272 (+0.234)	+0.249 (+0.218)	+0.188 (+0.163)	+0.208 (+0.183)
	2024, cohort 3 (development)	+0.317 (+0.251)	+0.287 (+0.230)	+0.226 (+0.165)	+0.246 (+0.188)
	2022, cohort 4 (confirmation)	+0.275 (+0.233)	+0.249 (+0.212)	+0.197 (+0.160)	+0.217 (+0.181)
H4, record recovered, 3b minus record only	2023 (development)	−0.088 (−0.113)	−0.071 (−0.089)	−0.027 (−0.039)	−0.032 (−0.043)
	2024, cohort 3 (development)	−0.094 (−0.113)	−0.075 (−0.089)	−0.027 (−0.039)	−0.034 (−0.045)
	2022, cohort 4 (confirmation)	−0.082 (−0.099)	−0.068 (−0.081)	−0.026 (−0.035)	−0.032 (−0.041)

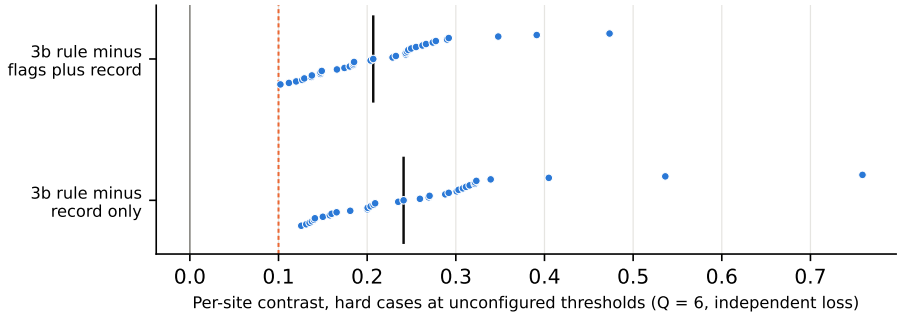


Figure 4: Per-site contrasts on the sealed cohort at six messages per day under independent loss (35 sites; the bar is the median; the dashed line is the pre-registered limit of the pooled contrast). Post hoc.

percentiles of the site contrasts were 0.117, 0.207, and 0.361. The record cost varied more between sites: at $Q = 6$ under independent loss the 5th percentile of the site differences was -0.160 and the median -0.062.

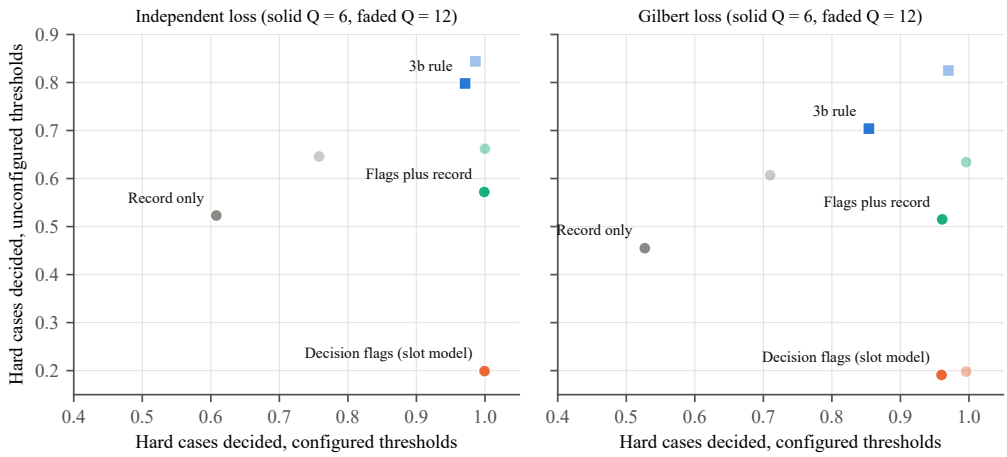


Figure 5: Hard cases decided at configured and at unconfigured thresholds on the sealed cohort (solid markers six messages per day, faded markers twelve). The flags decide almost every configured hard case; the 3b rule decides the most unconfigured ones.

5.5. Descriptive comparison

Table 5 and Fig. 5 show the pooled levels of the designs; these results carry no verdict.

- The flags-plus-record design decided 0.999 of hard cases at configured thresholds and 0.572 at unconfigured thresholds; the 3b rule decided 0.971 and 0.798. Flags that do not use the record (slot model) decided 0.199 at unconfigured thresholds.
- The record alone recovered the whole record on 0.842 of days and the 3b rule on 0.760; at $Q = 12$ the figures were 0.964 and 0.937 (Table 6).
- The slot model, run as a cross-check, reproduced the byte frame within 0.001 for the record alone and the flags-plus-record design and was pessimistic for the 3b rule by 0.027 to 0.036 in whole-record recovery, because it assigns the witness group whole slots.

Table 7 splits the contrasts by query kind, because the pooled hypotheses weight each kind by its number of hard cases.

- At unconfigured thresholds the 3b rule exceeded the flags-plus-record design for every kind under both loss models, with every interval above zero: by 0.106 (pH) to 0.311 (conductance change) under independent loss. No single kind drove the pooled result.
- At configured thresholds the flags-plus-record design exceeded the 3b rule for

Table 5: Pooled levels at six messages per day on the sealed cohort (record recovered, hard cases at configured (conf.) thresholds, hard cases at unconfigured (unconf.) thresholds, and all days at configured thresholds).

Design	Record	Hard, conf.	Hard, unconf.	All, conf.
Independent loss				
Record only	0.842	0.608	0.523	0.857
3b rule	0.760	0.971	0.798	0.882
Flags plus record	0.842	0.999	0.572	0.999
Decision flags (slot model)	0.842	0.999	0.199	0.999
Gilbert loss				
Record only	0.776	0.527	0.455	0.767
3b rule	0.707	0.854	0.704	0.793
Flags plus record	0.776	0.961	0.515	0.963
Decision flags (slot model)	0.775	0.960	0.191	0.960

Table 6: Pooled levels at twelve messages per day (same columns as Table 5).

Design	Record	Hard, conf.	Hard, unconf.	All, conf.
Independent loss				
Record only	0.964	0.758	0.646	0.954
3b rule	0.937	0.986	0.844	0.976
Flags plus record	0.964	1.000	0.662	1.000
Decision flags (slot model)	0.963	1.000	0.199	1.000
Gilbert loss				
Record only	0.939	0.710	0.607	0.924
3b rule	0.907	0.970	0.825	0.948
Flags plus record	0.939	0.996	0.634	0.997
Decision flags (slot model)	0.939	0.996	0.198	0.996

every kind. The largest gap was for the 7DADM under Gilbert loss, where a seven-day window needs an informative day in each of seven blocks and a burst of losses removes the witnesses that the flags do not need.

Table 8 splits the levels by region. The regions have one to five sites; therefore, the figures carry no interval and no verdict. The 3b rule decided the most unconfigured hard cases in every region. Its record cost relative to the record alone ranged from 0.015 in the Gulf and lower Mississippi region (two sites, where the record alone recovered 0.988 of days) to 0.129 in the Mid-Atlantic and Appalachia

Table 7: Contrasts at six messages per day by query kind, with 95% site-clustered intervals: the 3b rule minus flags plus record (F+R) and minus the record alone at unconfigured (unconf.) thresholds, and the 3b rule minus F+R at configured (conf.) thresholds.

Kind	3b − F+R, unconf.	3b − record, unconf.	3b − F+R, conf.
Independent loss, $Q = 6$			
Dissolved oxygen	+0.162 (+0.113 to +0.216)	+0.217 (+0.138 to +0.316)	−0.036 (−0.059 to −0.015)
pH	+0.106 (+0.066 to +0.151)	+0.127 (+0.081 to +0.178)	−0.028 (−0.055 to −0.011)
Conductance change	+0.311 (+0.279 to +0.341)	+0.368 (+0.318 to +0.424)	−0.020 (−0.024 to −0.016)
7DADM window	+0.258 (+0.216 to +0.296)	+0.335 (+0.284 to +0.381)	−0.043 (−0.057 to −0.030)
Gilbert loss, $Q = 6$			
Dissolved oxygen	+0.145 (+0.099 to +0.195)	+0.206 (+0.136 to +0.298)	−0.063 (−0.087 to −0.040)
pH	+0.095 (+0.061 to +0.134)	+0.116 (+0.075 to +0.162)	−0.071 (−0.096 to −0.054)
Conductance change	+0.270 (+0.241 to +0.296)	+0.335 (+0.290 to +0.383)	−0.065 (−0.069 to −0.060)
7DADM window	+0.163 (+0.130 to +0.196)	+0.289 (+0.255 to +0.320)	−0.289 (−0.309 to −0.268)

region (independent loss).

Table 8: Levels at six messages per day by region (hard cases decided at unconfigured thresholds, then record recovered, separated by a bar; number of sites in brackets; descriptive).

Region	Record only	3b rule	Flags plus record
Independent loss, $Q = 6$ (hard cases decided at unconfigured thresholds | record recovered)			
Gulf and lower Mississippi (2)	0.344 | 0.988	0.552 | 0.973	0.347 | 0.988
Mid-Atlantic and Appalachia (5)	0.522 | 0.783	0.755 | 0.654	0.560 | 0.783
Mountain West (5)	0.542 | 0.893	0.818 | 0.816	0.572 | 0.893
Northeast and New England (1)	0.381 | 0.881	0.581 | 0.823	0.397 | 0.881
Ohio valley and Great Lakes (5)	0.554 | 0.788	0.849 | 0.673	0.614 | 0.788
Pacific coast (5)	0.478 | 0.941	0.718 | 0.891	0.497 | 0.941
Southeast (5)	0.605 | 0.815	0.826 | 0.711	0.644 | 0.815
Texas (2)	0.546 | 0.899	0.832 | 0.828	0.586 | 0.899
Upper Midwest (5)	0.371 | 0.741	0.841 | 0.712	0.512 | 0.741
Gilbert loss, $Q = 6$ (hard cases decided at unconfigured thresholds | record recovered)			
Gulf and lower Mississippi (2)	0.292 | 0.927	0.477 | 0.903	0.306 | 0.927
Mid-Atlantic and Appalachia (5)	0.449 | 0.716	0.658 | 0.617	0.498 | 0.716
Mountain West (5)	0.451 | 0.818	0.704 | 0.759	0.496 | 0.818
Northeast and New England (1)	0.321 | 0.820	0.503 | 0.767	0.338 | 0.820
Ohio valley and Great Lakes (5)	0.494 | 0.731	0.757 | 0.636	0.567 | 0.731
Pacific coast (5)	0.423 | 0.867	0.647 | 0.826	0.445 | 0.867
Southeast (5)	0.536 | 0.749	0.736 | 0.660	0.585 | 0.749
Texas (2)	0.485 | 0.819	0.748 | 0.751	0.538 | 0.819
Upper Midwest (5)	0.317 | 0.690	0.738 | 0.657	0.464 | 0.690

5.6. Integrity under a forged message

The integrity study forced one arrived message per day to be forged: one to three bytes of its header or body were changed at random and its CRC-24 was recomputed; the receiver therefore accepted it as valid. Its results are descriptive (Table 9); the corresponding hypothesis H7 is an analytic bound.

- The receiver rejected the block in 0.80 of forged days at $Q = 6$ and 0.91 at $Q = 12$.
- False certificates (a reading certified outside its tolerance, or a witness code that differs from the truth) occurred only on days on which no redundant message arrived: exactly the needed number of messages (a false share of 0.16 to 0.22 on such days), fewer than needed, or uncoded oversize days. With more messages than needed the share was 0.000, as corroboration predicts.
- A wrong decision on the day followed a forgery on 0.019 (record only) and 0.022 (3b rule) of forged days at $Q = 6$, and on 0.002 at $Q = 12$. No decision-layer error occurred.
- Multiplied by the probability that a corrupted message passes the CRC-24 at the assumed bit-error rate of 10^{-3}, the expected number of days with a false certificate was at most 0.0035 per million days (upper bound, 3b rule, $Q = 6$); the ceiling for any receiver is 0.042 at $Q = 6$ and 0.084 at $Q = 12$.

Table 9: Integrity given one forged message that passes the CRC-24 (shares of forged days): block rejected; false-certificate share by class of day (more, exactly, or fewer messages than needed, and uncoded oversize days); and days with a wrong decision with their 95% interval.

Design	Q	Rejected	More	Exactly	Fewer	Uncoded	Wrong decision
Record only	6	0.805	0.000	0.214	0.146	0.171	0.0187 (0.0152 to 0.0226)
Record only	12	0.909	0.000	0.164	0.112	0.154	0.0017 (0.0010 to 0.0026)
3b rule	6	0.799	0.000	0.216	0.165	0.174	0.0219 (0.0184 to 0.0255)
3b rule	12	0.908	0.000	0.172	0.109	0.113	0.0024 (0.0016 to 0.0033)

5.7. Exploratory analyses after the confirmation (post hoc)

The analyses in this subsection were run after the confirmation, and none of them is a test. They reuse the frozen decoder and decision code at byte level, with loss draws shared by the three designs, and were run on the two development years first and on the sealed cohort second. Values are for the sealed cohort at six messages per day under independent loss 0.3 unless stated; the full tables are in Appendix D (Tables D.1, D.2, D.3, and D.4). No certain decision was wrong in any of these runs, except where a wrong reporting resolution was imposed on

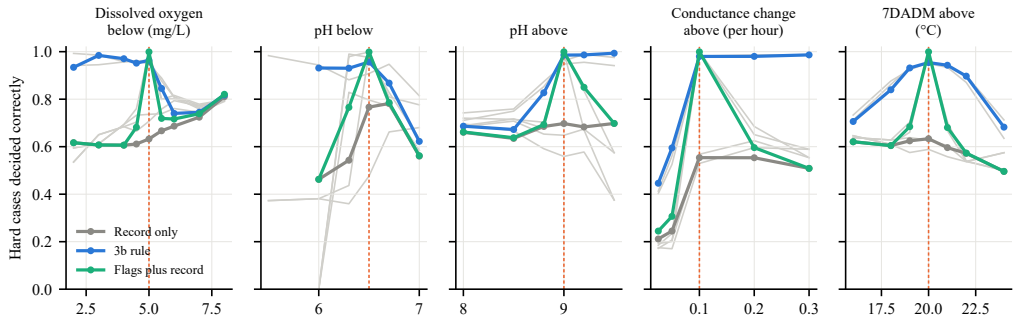


Figure 6: Hard cases decided correctly as the threshold moves away from the configured one (dashed), at six messages per day under independent loss 0.3; the sealed cohort in colour, the two development years in grey. Post hoc.

purpose. A first version of the threshold sweep reported 215 wrong 7DADM decisions at 22 °C on one development site; they came from the evaluation itself, which compared a seven-day mean that is exactly 22.0 in decimal arithmetic but 22.000000000000004 in floating point, and the certificate was right. Computed quantities are now rounded before the truth is evaluated.

Threshold sweep: Fig. 6 shows the share of hard cases decided correctly as the threshold moves away from the configured one, with a hard case defined at each threshold by the bands of the study. The flags-plus-record design decided 0.999 of hard cases at each configured threshold and fell to the level of the record alone beside it, because a configured answer implies little about another threshold. The 3b rule degraded in one direction only. Toward more extreme thresholds it stayed high: dissolved oxygen below 2 to 4.5 mg/L 0.934 to 0.984, pH below 6.0 and 6.3 about 0.93, pH above 9.2 and 9.5 0.986 and 0.993, and a conductance change above 20% and 30% 0.980 and 0.986 (record alone 0.46 to 0.70). Toward less extreme thresholds it fell to the level of the other designs: dissolved oxygen below 6 and 7 mg/L 0.740 and 0.745 (record alone 0.687 and 0.724), pH above 8.0 0.686 (0.661), and a conductance change above 2.5% 0.446 (0.211). The reason is the targeting rule: a day that is hard at a more extreme threshold is also near or across the configured one; therefore, its witnesses had the protected columns; a day that is hard only at a less extreme threshold was easy at the configured one, and its witnesses were only hosted. For the 7DADM, whose daily maxima are carried on hard days on either side, the 3b rule decided 0.840 to 0.954 between 18 and 22 °C and 0.706 and 0.682 at 16 and 24 °C. The two development years showed the same shapes (grey lines).

Loss sweep: Across independent loss of 0.1, 0.3, and 0.5, Gilbert loss of 0.3 with mean bursts of three and six messages, and Gilbert loss of 0.5 with bursts of three, at six and twelve messages per day, the contrast of the 3b rule with flags plus record at the unconfigured thresholds of the study stayed between 0.139 and 0.248 on the three years, with every lower bound at least 0.114; the contrast with the record alone stayed between 0.174 and 0.356. Longer bursts reduced the gain most (on the cohort 0.147 at six messages per day with bursts of six, against 0.191 with bursts of three). The cost in whole-record recovery grew with loss, from 0.053 at loss 0.1 to 0.096 at loss 0.5 on the cohort (six messages per day, independent loss), and stayed at most 0.096 in every condition on the cohort.

Record utility: On days whose whole record was not recovered, a substantial part of each day was still certified within tolerance: 0.416 of the readings for the record alone and 0.429 for the 3b rule at loss 0.3, and 0.28 to 0.29 at loss 0.5. Over all days, the share of readings certified was 0.908 for the record alone and 0.862 for the 3b rule. A failed whole-record recovery is therefore not a lost day, and the cost of the witnesses in certified readings (0.046 of readings) is about half of their cost in whole-record recovery.

Assumption R: With R off, the share of correct certain decisions fell by 0.003 to 0.006 on the three years, and no decision was wrong. With a deliberately wrong resolution (ten times too coarse, or 0.1 where none had been inferred), between 24,031 and 43,247 of about 1.8 million decisions per year and design were certain but wrong (1.3% to 2.3%). The assumption is therefore cheap to drop and dangerous to set wrongly: in deployment the resolution must be a declared, verified field of the site profile, not a value inferred from the record, and a site without a verified resolution should run with R off.

Baselines on the final frame: coded send-on-delta and the record at half the tolerance were compared with the record alone and the 3b rule in the slot model with four value slots per message (the frame of the confirmation; send-on-delta has no byte-level frame), on the sealed cohort (Table D.4). The slot model understates the 3b rule (by about 0.03 in whole-record recovery). Send-on-delta was the weakest design in every condition: at six messages per day under independent loss it recovered the whole record on 0.449 of days and decided 0.313 of hard cases at configured and 0.302 at unconfigured thresholds, against 0.842, 0.612, and 0.525 for the record alone. The record at half the tolerance recovered the whole record less often than the record alone in every condition (0.592 against 0.842 at six messages per day); at twelve messages per day it decided more hard cases than the record alone (0.731 against 0.647 at unconfigured thresholds, independent loss) but fewer than the 3b rule (0.820), and at six messages per day it decided fewer

than the record alone. The 3b rule decided the most hard cases at both kinds of threshold in every condition (for example 0.974 and 0.777 at six messages per day), at its cost in whole-record recovery. Neither alternative way of spending the bytes reached the decisions of the 3b rule, and neither preserved the record better than the record alone.

6. Discussion

6.1. What the evidence supports

The second confirmation supports a specific and bounded claim. On 35 rivers that no earlier round had used, measured at byte level, spending bytes on witnesses of hard cases decided 0.18 to 0.23 more hard cases at unconfigured thresholds than a node that sends the configured answers and also uses its record (H2), and 0.20 to 0.28 more than the record alone (H3), at a cost of 0.03 to 0.08 in the share of days whose whole record was recovered (H4). No certain decision was wrong (H1). The study does not claim that witnesses are the best way to decide thresholds that are fixed when the node is configured: the flags were better there in development and in both confirmations (H6).

The practical reading is a choice that depends on how fixed the thresholds are:

- If the thresholds are known and stable, a node that sends the configured answers with its record decides nearly every configured hard case (0.999 under independent loss at six messages per day) at no cost in record recovery, because on this frame the four flag bits fit in spare bits.
- If thresholds may change, or if the agency needs to decide thresholds it did not configure, the targeted witness design decides the most hard cases at unconfigured thresholds (0.798 against 0.572 for flags plus record and 0.523 for the record alone), at a cost of about 0.08 in whole-record recovery at six messages per day and 0.03 at twelve.

This is a claim about use, not a measured benefit in the field: the unconfigured threshold sets were chosen by the analyst, and whether thresholds change in practice was not studied.

The exploratory threshold sweep (Section 5.7) defines where the witness design helps. It helps for thresholds on the more extreme side of the configured one, and hardly at all on the other side. A worked use case on the sealed cohort illustrates the difference (six messages per day, independent loss 0.3). A monitoring programme configures its nodes for dissolved oxygen below 5 mg/L. Two years later the criterion for a reach is revised.

- Revised to a stricter low-oxygen criterion of 4 mg/L, for example to assess days of acute stress retrospectively: on the hard days the witness design decides 0.970 correctly, against 0.606 for flags plus record and for the record alone. The configured flag says little about 4 mg/L, but the received daily minimum settles it.
- Revised to 6 mg/L, for example for a cold-water designated use: the witness design decides 0.740, flags plus record 0.716, and the record alone 0.687. The days that matter at 6 mg/L were easy at 5 mg/L; therefore, their witnesses were not protected, and no design decides them well.

Two kinds of use should be kept apart. For compliance with a fixed criterion that carries legal consequences, the configured flags are the better design, and an undetermined answer should trigger a request for the stored record at the next service visit. For retrospective analysis, model calibration, threshold sensitivity studies, and the comparison of candidate criteria without reconfiguring remote nodes, the witness design keeps more decisions available, in particular for criteria that are stricter than the configured ones; an undetermined answer is then an honest statement that the bytes do not settle the question. A node intended for such use should be configured at the least extreme criterion of interest, because the witness design protects decisions on the more extreme side of its configured thresholds.

The reporting resolution (assumption R) needs an operational rule. The exploratory runs showed it cheap to drop (0.3 to 0.6 percentage points of decisions) and dangerous to set wrongly (1.3% to 2.3% of decisions certain but wrong with a resolution ten times too coarse). In this study the resolution was inferred from the record itself; in deployment it should be a declared field of the site profile, taken from the instrument's specified reporting precision and checked against the logger configuration, and a site without a verified resolution should run with R off.

6.2. What changed between the two confirmations, and why it matters

The first confirmation failed its integrity hypothesis and, read critically, overstated the benefit of witnesses: its flags comparator discarded the record; therefore, the contrast against flags measured the record as much as the witnesses. The revision replaced the comparator by one that uses the record, moved every hypothesis to the byte level, and lengthened the check. The benefit over the fair comparator (H2) is smaller than the earlier contrast against flags (0.53 at six messages per day under independent loss) but it is the contrast that isolates the witnesses, and it held on two development years and on the sealed cohort.

The design that was confirmed failed its own adoption criterion, and the confirmation does not change that. Table 10 lists the order of events. The adoption rule, written before the comparison it governed, required a gain of at least 0.10 over the

fair comparator (lower interval bound) and a pooled point cost in whole-record recovery of at most 0.08 against the record alone. On the CRC-24 frame the point cost at six messages per day under independent loss was 0.088 at byte level in development; the criterion therefore failed. The design was adopted because no other design met the benefit clause at a smaller cost, and its cost was then evaluated by a new hypothesis (H4) with looser limits, written before the rehearsals and the cohort. H4 is a later evaluation of the adopted design, not a retrospective satisfaction of the adoption rule: in the cohort the point cost at six messages per day under independent loss was 0.082, again beyond 0.08; at twelve messages per day it was 0.026 to 0.032 and met the rule. H2 and H3 are confirmatory evidence for the frozen design whatever the route by which it was chosen, because the cohort was new, the design and the limits were frozen before the data were downloaded, and the benefit hypotheses were not changed after the rehearsals.

The first argument for the adoption, that the cost was small against the fair comparator, rested on a modelling error that an internal reading found before the freeze: the flags had been charged a record value per message that the frame does not need. The argument was withdrawn, and the hypothesis built on it was dropped. The error did not affect the benefit hypotheses.

Table 10: Order of events between the two confirmations (all dates 2026).

Date	Event
9 November	First confirmation (version 2, 31 sites, 2024) run; its integrity hypothesis failed. An internal review found the flags comparator discarded the record
10 November	Version 3 planned; fair comparator added; adoption rule written before the comparison (gain at least 0.10, lower bound; point cost at most 0.08)
10 November	Integrity study on 2023 data selected the CRC-24 frame by a rule written before it ran
10 November	On the CRC-24 frame the 3b rule missed the cost clause (0.088 at byte level, six messages per day, independent loss); adopted after the fact
10 November	Draft pre-registration with H4 (cost limits -0.15 and -0.10) written; first rehearsal on the 2024 cohort, now development data
10 November	Second internal reading found the flag-capacity error; comparator corrected, H5 dropped, H7 relabelled, tests moved to byte level; limits unchanged; both rehearsals repeated
11 November	Plan and code frozen (tag and freeze record committed); 2022 cohort downloaded and sealed; one confirmatory run

6.3. Three different guarantees

The study provides three guarantees of different strength, and they should not be read as one:

- Erasure robustness (proved): whatever subset of messages arrives, and in whatever order, every certain answer is correct under the conditions of Proposition 1. Lost messages can only make answers undetermined.
- Detection of accidental corruption (conditional): a 24-bit check lets a corrupted message through with probability 2^{-24} under a model of independent random bit errors. At an assumed bit-error rate of 10^{-3} (a planning value, not a measurement of any satellite service) the expected rate of days with a false certificate was at most 0.0035 per million days, and no receiver could exceed 0.042 (six messages per day) or 0.084 (twelve). The figure is the expected number of arrived messages times the probability that one is corrupted times the probability that the corruption passes the check, times the measured share of false certificates given such a message; it is conservative because it pairs the mean number of arrived messages with the mean share. Burst errors longer than 24 bits, a faulty radio that repeats a frame, or a different error statistic would change it.
- Adversarial integrity (not provided): a cyclic redundancy check is not authentication. Anyone who can inject messages can compute a valid check, and the integrity study shows what one such message can do: given a forged message that passes the check, 0.11 to 0.22 of the days on which no surplus message arrived received a false certificate, and about 2% of forged days at six messages per day received a wrong decision. A deployment that must resist deliberate forgery needs a message authentication code, which costs payload bytes and was not studied.

6.4. Limits

- Simulation only: the evidence is trace-driven replay of USGS records through modelled loss. No satellite hardware, no measured channel error statistics, and no microcontroller measurements were used.
- Scope: 35 rivers of the United States in one year (2022), with one or two sites in three of the nine regions; nothing is claimed for other regions, years, or deployments.
- What was tested: three hypotheses could fail. H1 cannot fail through assumption R, H8 compares two implementations of the same code, H6 holds largely by construction, and H7 cannot exceed its ceiling.
- Assumption R: the reporting resolution was inferred from the analysed record for 26 of the 35 sites. H1 is therefore a test of the code, conditional on R; the

post-hoc runs with R off and with a wrong resolution show what each choice costs.

- Comparator: the flags-plus-record design takes its configured answers from the true record whenever a message arrives, which makes it an idealised node; its flag field is assumed to occupy four of the 16 spare bits, and the frame does not implement it.
- Comparisons outside the confirmation: send-on-delta and the tightened tolerance were compared with the final design only post hoc and in the slot model; the control for targeting, the exact allocator, and the hybrids were compared only in development and in the first confirmation.
- Censored codes: a forged record value with a censored code passes the sanity rule and removes a reading from decisions.
- Analyst choices: the alternative threshold sets, the hard-case bands, the 7DADM band, and the unknown-history rule were chosen by the analyst and were seen on every development cohort. The post-hoc threshold sweep shows that the benefit of the witnesses depends on the direction of the new threshold relative to the configured one.
- Loss models: independent and Gilbert loss with probabilities 0.1 to 0.5 and mean bursts of three or six messages (post hoc for all but 0.3 with bursts of three); no measured channel, no delay or reordering beyond what the frame tolerates by design, and no loss rate that varies over the day.
- Truth: truth statements refer to the loader's record, which includes readings bridged over gaps of at most two readings, and not to the river.
- Witness decisions are a lower bound on what witnesses could decide, because the box rule is not exact with witnesses (Proposition 2).
- Run: the confirmatory run was interrupted once by a restart of the computer and resumed with the same frozen code (Section 4.3).

7. Conclusion

This study changed the question for river water-quality telemetry over a one-way satellite link from how to deliver the whole record to what the server can soundly decide from what arrives. A certificate language turns each piece of received information into an interval and answers threshold, extremum, conductance-change, and rolling-mean queries with exceeded, not exceeded, or undetermined; the answers are sound when the intervals are valid, exact only without witnesses, and limited by the tolerance for readings that lie near a threshold.

The method was confirmed twice on sealed cohorts. The first confirmation failed its integrity hypothesis and used a comparator that discarded the record; the revised method was confirmed on 35 sealed rivers at byte level, where all three hypotheses that could fail held. No certain decision was wrong. On thresholds that nobody configured, targeted witnesses decided 0.18 to 0.23 more hard cases than a node sending the configured answers with its record, and 0.20 to 0.28 more than the record alone, at a cost of 0.03 to 0.08 in whole-record recovery. The design was adopted although that cost exceeded its own written adoption limit at six messages per day, and the paper reports it as such. Nodes that send the configured answers remained better on the thresholds they encode.

The claim of this study is generality and soundness, not the best decisions for known thresholds, and the evidence is trace-driven simulation. Two steps would move it toward practice: a measurement of real channel loss and error statistics, and an implementation of the frame on a satellite IoT module. The software, the frozen plans, and every result file are in the project repository.

Software and data availability

- Name: river-wq-satellite-records, version v3.0.0.
- Developers: Stephen Kwaku Azongo and Kofi Sarpong Adu-Manu, Department of Computer Science, University of Ghana.
- Contact: the corresponding author.
- Year first available: 2026.
- Language and requirements: Python 3.13 (packages listed in requirements.txt); a standard personal computer with 4 GB of memory runs every script.
- Size: about 17,000 lines of analysis code and 3,200 lines of tests.
- Licence: MIT for the code; CC BY 4.0 for the derived data and results.
- Repository: https://github.com/steveAzo/river-wq-satellite-records; archive of version v3.0.0, which accompanies this article: https://doi.org/10.5281/zenodo.24163716 (all versions: https://doi.org/10.5281/zenodo.23617664).
- Data: the USGS records were obtained from the National Water Information System and are in the public domain; the cleaned records, the sealed cohort with its checksum file, the frozen plans, and every result file are in the repository; the repository README lists the commands that reproduce each result. The Weija records belong to the deployment team and are available from the corresponding author on reasonable request.

CRediT authorship contribution statement

Stephen Kwaku Azongo: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. Kofi Sarpong Adu-Manu: Conceptualization, Methodology, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic), an AI assistant, to help write and revise the analysis code and to draft and edit the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Appendix A. Proofs of Propositions 1 and 2

Setting: A channel-day has readings $x(0),\dots,x(n-1)$. A reading is censored when it lies outside the quantiser range; V is the set of uncensored readings, and every query is defined on V. The server holds for every reading an interval $[\mathrm{lo}(t),\mathrm{hi}(t)]$, a set E of readings known to be censored, and optional global bounds g_{lo}, g_{hi} (from witnesses); it treats S, the complement of E, as uncensored. The knowledge is valid on V when $x(t) \in [\mathrm{lo}(t),\mathrm{hi}(t)]$ for all $t \in V$, $g_{lo} \le x(t) \le g_{hi}$ for all $t \in V$, and $E \cap V = \emptyset$; then $V \subseteq S$. All rules use the effective intervals of Eq. (3), written here lo' and hi', which are valid on V when the knowledge is. Under assumption R both ends are moved inward to the reporting grid; because every reading in V lies on the grid, the moved interval still contains it, and the arguments below hold for the moved intervals unchanged.

Lemma 1 (tolerance interval): For $r < 1$ and $x \geq 0$, $\{x : |x - \hat{x}| \leq \max(a, rx)\}$ is the interval of Eq. (2), with the tolerance of Eq. (1). Proof: the set is the union of $\{|x - \hat{x}| \leq a\}$ and $\{|x - \hat{x}| \leq rx\}$; both contain $\hat{x}$, the first is $[\hat{x} - a, \hat{x} + a]$, and the second, for $x \geq 0$, is $[\hat{x}/(1 + r), \hat{x}/(1 - r)]$. Two intervals that share a point have an interval as union, with the extreme endpoints.

Lemma C (cell): Nearest rounding maps x to code c only if $|x - \tilde{x}(c)| \leq s/2$, where $\tilde{x}(c)$ is the dequantised value and s the step (on the logarithmic grid of conductance, a factor of $10^{s/2}$).

Proposition 1, part 1 (answers that hold over any superset): Let the knowledge be valid on $V \subseteq S$. (i) Threshold rule of Eq. (4), "some reading below θ", answer 0: every $t \in S$ has $\mathrm{lo}'(t) \geq \theta$; since $V \subseteq S$ and each $x(t) \geq \mathrm{lo}'(t)$ on V, no reading of V is below θ. The rule "some reading above θ" is symmetric. (ii) Conductance rule, answer 0: every pair of S at the lag is certainly within the fraction; the pairs of V are a subset, and each pair's ratio lies in the image of its intervals, Eq. (6) (exact for a box); therefore, every pair of V is within the fraction. (iii) Day extrema of Eq. (5). Day minimum, lower end: $\min_S \mathrm{lo}' \leq \min_V \mathrm{lo}' \leq \min_V x$. Day maximum, upper end: $\max_S \mathrm{hi}' \geq \max_V \mathrm{hi}' \geq \max_V x$. None of these uses $S \setminus V$.

Proposition 1, part 2 (answers that need condition B): Condition B: for every $t \in S \setminus V$ there are $u, v \in V$ with $\mathrm{lo}'(t) \leq x(u)$ and $\mathrm{hi}'(t) \geq x(v)$. (i) "Some reading below θ", answer 1: some $t \in S$ has $\mathrm{hi}'(t) < \theta$. If $t \in V$, then $x(t) \leq \mathrm{hi}'(t) < \theta$. If $t \in S \setminus V$, then $x(v) \leq \mathrm{hi}'(t) < \theta$ for some $v \in V$. In both cases a reading of V is below θ. "Above" is symmetric. (ii) Day minimum, upper end: $\min_S \mathrm{hi}'$ is attained at some t. If $t \in V$, then $\min_V x \leq x(t) \leq \mathrm{hi}'(t)$; if $t \in S \setminus V$, then $\min_V x \leq x(v) \leq \mathrm{hi}'(t)$. Hence $\min_V x \leq \min_S \mathrm{hi}'$, and with part 1 the minimum lies in its interval. The maximum is symmetric. (iii) 7DADM, Eq. (7): each daily-maximum interval contains the true daily maximum by (ii) and part 1 (a day with no information contributes the base range, which contains it as well); the mean of seven values lies between the means of the lower and of the upper ends; therefore, answer 1 (lower mean above the threshold) and answer 0 (upper mean at most the threshold) are correct.

Proposition 1, part 3 (conductance, answer 1): If no conductance reading is in $S \setminus V$, then $S = V$ on that channel, the knowledge is valid on every reading the rule uses, and the rule is the exact image of the box under "all pairs within the fraction" along each chain (a reachable interval propagated by $[0.9a, 1.1b]$ and intersected with the next interval); an empty reachable set means no assignment of the true readings avoids a change; therefore, the answer 1 is correct. With an unrecognised censored conductance reading the chain would pass through a reading outside the domain, and part 3 makes no claim.

Lemma 2 (when condition B holds): Let $t \in S \setminus V$ lie in a certified span whose bounding kept values have uncensored codes c_p, c_q at readings $p, q \in V$. The reconstruction $\hat{x}(t)$ is the linear interpolation of $\tilde{x}_p = \tilde{x}(c_p)$ and $\tilde{x}_q = \tilde{x}(c_q)$ (or equals one of them at the ends of the day); therefore, $\min(\tilde{x}_p, \tilde{x}_q) \le \hat{x}(t) \le \max(\tilde{x}_p, \tilde{x}_q)$. By Lemma 1, $\mathrm{hi}(t) \ge \hat{x}(t) + a$ (linear parameters) or $\hat{x}(t)/(1-r)$ (conductance), and by Lemma C, $x(p) \le \tilde{x}_p + s/2$. The tolerance constants exceed half a step for every parameter (at least 0.2 against at most 0.0025 units; 3% against 0.07% for conductance); therefore, $\mathrm{hi}(t) \ge \min(x(p), x(q))$ and, symmetrically, $\mathrm{lo}(t) \le \max(x(p), x(q))$. Both bounds survive the effective intervals: a witness bound holds on V; therefore, $g_{lo} \le \min(x(p), x(q))$ and $g_{hi} \ge \max(x(p), x(q))$, and the grid move keeps the grid values $x(p), x(q)$ inside. Hence condition B holds with $u, v \in \{p, q\}$. If a bounding kept value has a censored code, its dequantised value is the edge of the range and the argument fails on that side only.

Proposition 2 (exactness without witnesses): Without witnesses, the set of records consistent with the knowledge is the box $\prod_t [\mathrm{lo}(t), \mathrm{hi}(t)]$ over S (taking $S = V$, as when no censored reading is unrecognised). For "some reading below θ": over the box, $\min_t x(t)$ ranges exactly over $[\min \mathrm{lo}, \min \mathrm{hi}]$ (each end attained by setting every reading to its lower or upper end). The query is true on every consistent record iff $\min \mathrm{hi} < \theta$, false on every one iff $\min \mathrm{lo} \ge \theta$, and takes both values otherwise; these are exactly the cases in which the rule answers 1, 0, and -1. "Above" and the extrema are symmetric, and the extremum ends are attained. With a witness, the consistent set is no longer a box: the witness also asserts that some reading lies in its cell. Counterexample: reading 1 in $[3, 10]$, reading 2 in $[4.5, 4.6]$, minimum witness in $[4.0, 4.01]$. Reading 2 cannot be the minimum; therefore, reading 1 lies in $[4.0, 4.01]$ and "some reading above 5" is certainly false, but the box rule (which uses only $x \ge g_{lo}$) answers -1. The rules are therefore sound (Proposition 1) but not exact with witnesses.

Appendix B. Sites of the sealed confirmation cohort

The 35 sites of the second confirmation were selected by metadata only (seed 20261113, at most five per region; the full selection log is in the repository). The days are four-parameter days with a record plan in the confirmation run (10,417 in total).

Table B.1: Sites of the sealed 2022 cohort by region.

Site	Name	State	Days
Gulf and lower Mississippi			
07178200	Bird Ck At State Highway 266 Near Catoosa, Ok	OK	100
07374000	Mississippi River At Baton Rouge, La	LA	324
Mid-Atlantic and Appalachia			
01480870	East Branch Brandywine Creek Below Downingtown, Pa	PA	265
01575900	Chiques Creek Near Marietta, Pa	PA	354
01645762	S F Little Difficult Run Above Mouth Nr Vienna, Va	VA	302
01646305	Dead Run At Whann Avenue Near Mclean, Va	VA	310
01649500	Northeast Branch Anacostia River At Riverdale, Md	MD	340
Mountain West			
09149500	Uncompahgre River At Delta, Co	CO	356
09180500	Colorado River Near Cisco, Ut	UT	307
12301933	Kootenai River Bl Libby Dam Nr Libby Mt	MT	357
13093383	Snake River At Pigeon Cove Nr Twin Falls, Id	ID	361
393259107194801	Colorado R Abv Roaring Fork R At Glenwood Spgs, Co	CO	299
Northeast and New England			
04249000	Oswego River At Lock 7, Oswego Ny	NY	343
Ohio valley and Great Lakes			
03292500	South Fork Beargrass Creek At Louisville, Ky	KY	340
03297900	Floyds Fork Near Pewee Valley, Ky	KY	241
03301900	Fern Creek At Old Bardstown Rd At Louisville, Ky	KY	312
03302060	Salt River At Mouth At West Point, Ky	KY	299
05576100	Lick Creek Near Woodside, Il	IL	303
Pacific coast			
11273400	San Joaquin R Ab Merced R Nr Newman Ca	CA	251
12100490	White River At R Street Near Auburn, Wa	WA	251
14181500	North Santiam River At Niagara, Or	OR	239
14202980	Scoggins Ck Blw Henry Hagg Lake, Nr Gaston, Or	OR	359
14316460	North Umpqua R At Soda Spgs, Nr Toketee Falls, Or	OR	358
Southeast			
0208062765	Roanoke River At Halifax, Nc	NC	349

Site	Name	State	Days
02203655	South River At Forrest Park Road, At Atlanta, Ga	GA	344
02336300	Peachtree Creek At Atlanta, Ga	GA	330
02336340	Nancy Creek At Johnson Ferry Rd, At Chamblee, Ga	GA	328
02336410	Nancy Creek At West Wesley Road, At Atlanta, Ga	GA	333
Texas			
08065350	Trinity Rv Nr Crockett, Tx	TX	264
08188060	San Antonio Rv At Sh 72 Nr Runge, Tx	TX	299
Upper Midwest			
04067500	Menominee River Near Mc Allister, Wi	WI	274
040851385	Fox River At Oil Tank Depot At Green Bay, Wi	WI	350
04087170	Milwaukee River At Mouth At Milwaukee, Wi	WI	148
05474500	Mississippi River At Keokuk, Ia	IA	116
07020850	Mississippi River At Cape Girardeau, Mo	MO	311

Appendix C. The first confirmation (version 2 of the method)

The first confirmation tested version 2 of the method (CRC-16 frame, slot-model comparisons, flags that do not use the record) on 31 sealed sites of 2024. It is reported here in full because it shaped the revision. Pairs of values are independent and Gilbert loss at six messages per day. Caveats found after that run: H7 and the configured and record parts of H9 hold by construction; the flags comparator of H5 discarded the record; flags were modelled at 4.75 values per message; and the forgery study of H8 changed body bytes only. After that run its cohort became development data.

Table C.1: Hypotheses of the first confirmation and their outcome on its 31 sealed sites.

ID	Statement	Limit	Result (95% interval)	Outcome
H1	No wrong decision in any certificate design; no frame round-trip mismatch	0 wrong; 0 mismatches	wrong decisions max 0.000; round trip: 0 mismatches in 30285 blocks	Supported

ID	Statement	Limit	Result (95% interval)	Outcome
H2	3b rule minus record only, hard cases at configured thresholds	lower bound at least 0.15	+0.318 (+0.234 to +0.417); +0.300 (+0.223 to +0.390)	Supported
H3	3b rule minus random-day layering, hard cases	lower bound at least 0.05	+0.141 (+0.105 to +0.174); +0.130 (+0.096 to +0.165)	Supported
H4	Record recovered, 3b rule minus record only	lower bound at least −0.12	−0.083 (−0.105 to −0.062); −0.071 (−0.091 to −0.053)	Supported
H5	Unanticipated thresholds, hard cases: 3b rule minus flags, and minus record only	both lower bounds at least 0.10	flags: +0.531 (+0.494 to +0.569); +0.441 (+0.411 to +0.472); record only: +0.260 (+0.191 to +0.344); +0.245 (+0.184 to +0.320)	Supported
H6	Neither send-on-delta nor tolerance x 0.5 dominates the 3b rule	no dominance	send-on-delta/independent: no; tolerance x 0.5/independent: no; send-on-delta/Gilbert: no; tolerance x 0.5/Gilbert: no	Supported
H7	Known limit: flags minus 3b rule, configured thresholds, independent loss	lower bound above 0	+0.043 (+0.025 to +0.068)	Supported
H8	False-certificate day share given one CRC-valid forgery	upper bounds at most 0.06, 0.01, 0.15, 0.02	record, Q = 6: 0.0463 (upper 0.0584, limit 0.06); record, Q = 12: 0.0061 (upper 0.0134, limit 0.01); layered, Q = 6: 0.1050 (upper 0.1197, limit 0.15); layered, Q = 12: 0.0126 (upper 0.0215, limit 0.02)	Not supported
H9	Hybrid 1 minus flags: unanticipated thresholds; configured thresholds; record recovered	alt lower bound at least 0.20; others at least −0.01	alt: +0.355 (+0.297 to +0.418); +0.294 (+0.248 to +0.345); configured: +0.000 (+0.000 to +0.000); +0.000 (+0.000 to +0.000); record: +0.000 (+0.000 to +0.000); +0.000 (+0.000 to +0.000)	Supported

Appendix D. Exploratory analyses after the confirmation (post hoc)

Run after the confirmation; none is a test. Byte level (CRC-24 frame, hardened receiver, decoded-only decisions), loss draws shared by the designs, 20 draws per day and condition, every analysed day of each year. Unconfigured thresholds are the two alternative sets of the study.

Table D.1: Loss sweep: contrasts at the unconfigured thresholds (hard cases) and in whole-record recovery (point estimate with the lower bound of the site-clustered 95% interval; ind. is independent loss, Gil. Gilbert loss with mean burst length b).

Year	Q	Loss	3b − flags plus record	3b − record only	Record recovered, 3b − record
2023 (dev.)	6	ind. 0.1	+0.189 (+0.162)	+0.212 (+0.179)	−0.061 (−0.083)
2023 (dev.)	6	ind. 0.3	+0.223 (+0.196)	+0.271 (+0.236)	−0.089 (−0.113)
2023 (dev.)	6	ind. 0.5	+0.218 (+0.194)	+0.311 (+0.279)	−0.094 (−0.113)
2023 (dev.)	6	Gil. 0.3, b 3	+0.183 (+0.160)	+0.247 (+0.219)	−0.071 (−0.089)
2023 (dev.)	6	Gil. 0.3, b 6	+0.139 (+0.117)	+0.199 (+0.175)	−0.056 (−0.071)
2023 (dev.)	6	Gil. 0.5, b 3	+0.162 (+0.141)	+0.255 (+0.228)	−0.080 (−0.096)
2023 (dev.)	12	ind. 0.1	+0.171 (+0.148)	+0.174 (+0.152)	−0.014 (−0.023)
2023 (dev.)	12	ind. 0.3	+0.178 (+0.154)	+0.186 (+0.162)	−0.027 (−0.039)
2023 (dev.)	12	ind. 0.5	+0.204 (+0.178)	+0.233 (+0.202)	−0.056 (−0.073)
2023 (dev.)	12	Gil. 0.3, b 3	+0.187 (+0.164)	+0.208 (+0.183)	−0.032 (−0.043)
2023 (dev.)	12	Gil. 0.3, b 6	+0.170 (+0.149)	+0.205 (+0.183)	−0.030 (−0.039)
2023 (dev.)	12	Gil. 0.5, b 3	+0.209 (+0.185)	+0.253 (+0.223)	−0.054 (−0.068)
2024 (dev.)	6	ind. 0.1	+0.211 (+0.162)	+0.250 (+0.179)	−0.059 (−0.077)
2024 (dev.)	6	ind. 0.3	+0.248 (+0.204)	+0.318 (+0.253)	−0.094 (−0.113)
2024 (dev.)	6	ind. 0.5	+0.235 (+0.204)	+0.356 (+0.304)	−0.108 (−0.127)
2024 (dev.)	6	Gil. 0.3, b 3	+0.202 (+0.166)	+0.288 (+0.233)	−0.076 (−0.090)
2024 (dev.)	6	Gil. 0.3, b 6	+0.151 (+0.114)	+0.231 (+0.178)	−0.059 (−0.071)
2024 (dev.)	6	Gil. 0.5, b 3	+0.170 (+0.142)	+0.290 (+0.241)	−0.090 (−0.104)
2024 (dev.)	12	ind. 0.1	+0.194 (+0.150)	+0.211 (+0.154)	−0.018 (−0.032)
2024 (dev.)	12	ind. 0.3	+0.202 (+0.156)	+0.225 (+0.164)	−0.028 (−0.041)
2024 (dev.)	12	ind. 0.5	+0.229 (+0.184)	+0.278 (+0.215)	−0.056 (−0.070)
2024 (dev.)	12	Gil. 0.3, b 3	+0.209 (+0.166)	+0.247 (+0.188)	−0.034 (−0.047)
2024 (dev.)	12	Gil. 0.3, b 6	+0.189 (+0.150)	+0.241 (+0.190)	−0.034 (−0.045)
2024 (dev.)	12	Gil. 0.5, b 3	+0.231 (+0.190)	+0.296 (+0.239)	−0.057 (−0.070)
2022 (cohort)	6	ind. 0.1	+0.191 (+0.161)	+0.215 (+0.171)	−0.053 (−0.066)
2022 (cohort)	6	ind. 0.3	+0.226 (+0.197)	+0.275 (+0.230)	−0.083 (−0.101)
2022 (cohort)	6	ind. 0.5	+0.227 (+0.202)	+0.316 (+0.274)	−0.096 (−0.112)
2022 (cohort)	6	Gil. 0.3, b 3	+0.191 (+0.167)	+0.252 (+0.213)	−0.067 (−0.079)
2022 (cohort)	6	Gil. 0.3, b 6	+0.147 (+0.125)	+0.202 (+0.167)	−0.052 (−0.061)
2022 (cohort)	6	Gil. 0.5, b 3	+0.174 (+0.153)	+0.261 (+0.224)	−0.080 (−0.093)
2022 (cohort)	12	ind. 0.1	+0.176 (+0.149)	+0.188 (+0.151)	−0.016 (−0.024)
2022 (cohort)	12	ind. 0.3	+0.182 (+0.154)	+0.197 (+0.158)	−0.026 (−0.035)
2022 (cohort)	12	ind. 0.5	+0.208 (+0.179)	+0.242 (+0.201)	−0.051 (−0.063)
2022 (cohort)	12	Gil. 0.3, b 3	+0.191 (+0.164)	+0.216 (+0.179)	−0.031 (−0.039)
2022 (cohort)	12	Gil. 0.3, b 6	+0.173 (+0.149)	+0.212 (+0.177)	−0.030 (−0.038)
2022 (cohort)	12	Gil. 0.5, b 3	+0.214 (+0.187)	+0.261 (+0.222)	−0.052 (−0.062)

Table D.2: Record utility: whole record recovered, share of readings certified within tolerance over all draws, and on draws whose whole record was not recovered.

Year	Q	Loss	Design	Whole record	Certified, all	Certified, not whole
2023 (dev.)	6	ind. 0.3	Record only	0.846	0.912	0.430
2023 (dev.)	6	ind. 0.3	3b rule	0.757	0.863	0.436
2023 (dev.)	6	ind. 0.3	Flags plus record	0.846	0.912	0.430
2023 (dev.)	6	ind. 0.5	Record only	0.660	0.759	0.291
2023 (dev.)	6	ind. 0.5	3b rule	0.566	0.696	0.300
2023 (dev.)	6	ind. 0.5	Flags plus record	0.660	0.759	0.291
2023 (dev.)	6	Gil. 0.3, b 3	Record only	0.780	0.842	0.283
2023 (dev.)	6	Gil. 0.3, b 3	3b rule	0.709	0.803	0.324
2023 (dev.)	6	Gil. 0.3, b 3	Flags plus record	0.780	0.842	0.283
2023 (dev.)	12	ind. 0.3	Record only	0.980	0.990	0.515
2023 (dev.)	12	ind. 0.3	3b rule	0.953	0.976	0.486
2023 (dev.)	12	ind. 0.3	Flags plus record	0.980	0.990	0.515
2023 (dev.)	12	ind. 0.5	Record only	0.928	0.951	0.314
2023 (dev.)	12	ind. 0.5	3b rule	0.873	0.913	0.313
2023 (dev.)	12	ind. 0.5	Flags plus record	0.928	0.951	0.314
2023 (dev.)	12	Gil. 0.3, b 3	Record only	0.954	0.967	0.297
2023 (dev.)	12	Gil. 0.3, b 3	3b rule	0.922	0.949	0.345
2023 (dev.)	12	Gil. 0.3, b 3	Flags plus record	0.954	0.967	0.297
2024 (dev.)	6	ind. 0.3	Record only	0.838	0.903	0.400
2024 (dev.)	6	ind. 0.3	3b rule	0.744	0.849	0.411
2024 (dev.)	6	ind. 0.3	Flags plus record	0.838	0.903	0.400
2024 (dev.)	6	ind. 0.5	Record only	0.651	0.747	0.275
2024 (dev.)	6	ind. 0.5	3b rule	0.542	0.675	0.289
2024 (dev.)	6	ind. 0.5	Flags plus record	0.651	0.747	0.275
2024 (dev.)	6	Gil. 0.3, b 3	Record only	0.768	0.831	0.269
2024 (dev.)	6	Gil. 0.3, b 3	3b rule	0.692	0.787	0.307
2024 (dev.)	6	Gil. 0.3, b 3	Flags plus record	0.768	0.831	0.269
2024 (dev.)	12	ind. 0.3	Record only	0.961	0.978	0.435
2024 (dev.)	12	ind. 0.3	3b rule	0.933	0.960	0.407
2024 (dev.)	12	ind. 0.3	Flags plus record	0.961	0.978	0.435
2024 (dev.)	12	ind. 0.5	Record only	0.913	0.939	0.298
2024 (dev.)	12	ind. 0.5	3b rule	0.858	0.898	0.284
2024 (dev.)	12	ind. 0.5	Flags plus record	0.913	0.939	0.298
2024 (dev.)	12	Gil. 0.3, b 3	Record only	0.936	0.956	0.313
2024 (dev.)	12	Gil. 0.3, b 3	3b rule	0.902	0.933	0.322
2024 (dev.)	12	Gil. 0.3, b 3	Flags plus record	0.936	0.956	0.313
2022 (cohort)	6	ind. 0.3	Record only	0.842	0.908	0.416
2022 (cohort)	6	ind. 0.3	3b rule	0.759	0.862	0.429
2022 (cohort)	6	ind. 0.3	Flags plus record	0.842	0.908	0.416
2022 (cohort)	6	ind. 0.5	Record only	0.663	0.758	0.282
2022 (cohort)	6	ind. 0.5	3b rule	0.568	0.694	0.292
2022 (cohort)	6	ind. 0.5	Flags plus record	0.663	0.758	0.282
2022 (cohort)	6	Gil. 0.3, b 3	Record only	0.776	0.839	0.281
2022 (cohort)	6	Gil. 0.3, b 3	3b rule	0.709	0.802	0.319
2022 (cohort)	6	Gil. 0.3, b 3	Flags plus record	0.776	0.839	0.281
2022 (cohort)	12	ind. 0.3	Record only	0.963	0.979	0.413
2022 (cohort)	12	ind. 0.3	3b rule	0.937	0.964	0.432
2022 (cohort)	12	ind. 0.3	Flags plus record	0.963	0.979	0.413
2022 (cohort)	12	ind. 0.5	Record only	0.917	0.941	0.287
2022 (cohort)	12	ind. 0.5	3b rule	0.866	0.905	0.293
2022 (cohort)	12	ind. 0.5	Flags plus record	0.917	0.941	0.287
2022 (cohort)	12	Gil. 0.3, b 3	Record only	0.939	0.957	0.300

Year	Q	Loss	Design	Whole record	Certified, all	Certified, not whole
2022 (cohort)	12	Gil. 0.3, b 3	3b rule	0.908	0.939	0.341
2022 (cohort)	12	Gil. 0.3, b 3	Flags plus record	0.939	0.957	0.300

Table D.3: Assumption R at six messages per day, independent loss 0.3, configured and alternative thresholds (dissolved oxygen, pH, conductance): share of decisions certain and correct, and certain but wrong, with the inferred resolution, with R off, and with a resolution ten times too coarse (0.1 where none was inferred).

Year	Design	Resolution	Correct	Wrong
2023 (dev.)	3b rule	inferred	0.8208	0 of 1,821,060
2023 (dev.)	3b rule	off	0.8174	0 of 1,821,060
2023 (dev.)	3b rule	wrong	0.8087	29,549 of 1,821,060
2023 (dev.)	Record only	inferred	0.7865	0 of 1,821,060
2023 (dev.)	Record only	off	0.7837	0 of 1,821,060
2023 (dev.)	Record only	wrong	0.7797	28,028 of 1,821,060
2024 (dev.)	3b rule	inferred	0.8194	0 of 1,817,100
2024 (dev.)	3b rule	off	0.8138	0 of 1,817,100
2024 (dev.)	3b rule	wrong	0.8091	28,704 of 1,817,100
2024 (dev.)	Record only	inferred	0.7741	0 of 1,817,100
2024 (dev.)	Record only	off	0.7698	0 of 1,817,100
2024 (dev.)	Record only	wrong	0.7721	24,031 of 1,817,100
2022 (cohort)	3b rule	inferred	0.8216	0 of 1,875,060
2022 (cohort)	3b rule	off	0.8171	0 of 1,875,060
2022 (cohort)	3b rule	wrong	0.8010	43,247 of 1,875,060
2022 (cohort)	Record only	inferred	0.7910	0 of 1,875,060
2022 (cohort)	Record only	off	0.7861	0 of 1,875,060
2022 (cohort)	Record only	wrong	0.7766	39,107 of 1,875,060

Table D.4: Baselines on the final frame in the slot model (four value slots per message): whole record recovered and hard cases decided at configured and unconfigured thresholds. The slot model is pessimistic for the 3b rule (whole slots for the witness group).

Year	Q	Loss	Design	Record	Hard, configured	Hard, unconfigured
2023 (dev.)	6	ind.	Record only	0.844	0.615	0.525
2023 (dev.)	6	ind.	3b rule (slot model)	0.718	0.964	0.762
2023 (dev.)	6	ind.	Tolerance x 0.5	0.561	0.427	0.476
2023 (dev.)	6	ind.	Send-on-delta, coded	0.548	0.359	0.369
2023 (dev.)	6	Gil.	Record only	0.780	0.523	0.449
2023 (dev.)	6	Gil.	3b rule (slot model)	0.673	0.847	0.671
2023 (dev.)	6	Gil.	Tolerance x 0.5	0.545	0.380	0.427
2023 (dev.)	6	Gil.	Send-on-delta, coded	0.531	0.311	0.325
2023 (dev.)	12	ind.	Record only	0.980	0.777	0.653
2023 (dev.)	12	ind.	3b rule (slot model)	0.919	0.981	0.806
2023 (dev.)	12	ind.	Tolerance x 0.5	0.868	0.756	0.744
2023 (dev.)	12	ind.	Send-on-delta, coded	0.868	0.592	0.579
2023 (dev.)	12	Gil.	Record only	0.954	0.725	0.612
2023 (dev.)	12	Gil.	3b rule (slot model)	0.890	0.965	0.788
2023 (dev.)	12	Gil.	Tolerance x 0.5	0.832	0.689	0.686
2023 (dev.)	12	Gil.	Send-on-delta, coded	0.832	0.530	0.526

Year	Q	Loss	Design	Record	Hard, configured	Hard, unconfigured
2022 (cohort)	6	ind.	Record only	0.842	0.612	0.525
2022 (cohort)	6	ind.	3b rule (slot model)	0.724	0.974	0.777
2022 (cohort)	6	ind.	Tolerance x 0.5	0.592	0.438	0.475
2022 (cohort)	6	ind.	Send-on-delta, coded	0.449	0.313	0.302
2022 (cohort)	6	Gil.	Record only	0.775	0.524	0.453
2022 (cohort)	6	Gil.	3b rule (slot model)	0.677	0.861	0.689
2022 (cohort)	6	Gil.	Tolerance x 0.5	0.571	0.402	0.438
2022 (cohort)	6	Gil.	Send-on-delta, coded	0.447	0.278	0.273
2022 (cohort)	12	ind.	Record only	0.963	0.758	0.647
2022 (cohort)	12	ind.	3b rule (slot model)	0.908	0.987	0.820
2022 (cohort)	12	ind.	Tolerance x 0.5	0.876	0.751	0.731
2022 (cohort)	12	ind.	Send-on-delta, coded	0.792	0.549	0.534
2022 (cohort)	12	Gil.	Record only	0.939	0.712	0.608
2022 (cohort)	12	Gil.	3b rule (slot model)	0.880	0.972	0.804
2022 (cohort)	12	Gil.	Tolerance x 0.5	0.838	0.686	0.673
2022 (cohort)	12	Gil.	Send-on-delta, coded	0.762	0.496	0.488

References

Adu-Manu, K.S., Katsriku, F.A., Abdulai, J.D., Engmann, F., 2020. Smart river monitoring using wireless sensor networks. Wireless Communications and Mobile Computing 2020, 1–19. doi:10.1155/2020/8897126.

Adu-Manu, K.S., Tapparello, C., Heinzelman, W., Katsriku, F.A., Abdulai, J.D., 2017. Water quality monitoring using wireless sensor networks: Current trends and future research directions. ACM Transactions on Sensor Networks 13, 1–41. doi:10.1145/3005719.

Afhamisis, M., Palattella, M.R., 2022. SALSA: A scheduling algorithm for LoRa to LEO satellites. IEEE Access 10, 11608–11615. doi:10.1109/access.2022.3146021.

Ahmedi, F., Ahmedi, L., 2022. Dataset on water quality monitoring from a wireless sensor network in a river in Kosovo. Data in Brief 44, 108486. doi:10.1016/j.dib.2022.108486.

Albanese, A., Blomer, J., Edmonds, J., Luby, M., Sudan, M., 1996. Priority encoding transmission. IEEE Transactions on Information Theory 42, 1737–1744. doi:10.1109/18.556670.

Alvarez, G., Fraire, J.A., Hassan, K.A., Cespedes, S., Pesch, D., 2022. Uplink transmission policies for LoRa-based direct-to-satellite IoT. IEEE Access 10, 72687–72701. doi:10.1109/access.2022.3189647.

Boehme, L., Lovell, P., Biuw, M., Roquet, F., Nicholson, J., Thorpe, S.E., Meredith, M.P., Fedak, M., 2009. Technical note: Animal-borne CTD-satellite relay data loggers for real-time oceanographic data collection. Ocean Science 5, 685–695. doi:10.5194/os-5-685-2009.

Boucheron, S., Salamatian, M.R., 2000. About priority encoding transmission. IEEE Transactions on Information Theory 46, 699–705.

Brito, J., Katsis, Y., Demirkaya, K., Lin, C., Boursier, E., Papakonstantinou, Y., 2017. Efficient approximate query answering over sensor data with deterministic error guarantees. ArXiv preprint arXiv:1707.01414.

Campbell, J.L., Rustad, L.E., Porter, J.H., Taylor, J.R., Dereszynski, E.W., Shanley, J.B., Gries, C., Henshaw, D.L., Martin, M.E., Sheldon, W.M., Boose, E.R., 2013. Quantity is nothing without quality: Automated QA/QC for streaming environmental sensor data. BioScience 63, 574–585. doi:10.1525/bio.2013.63.7.10.

Capez, G.M., Henn, S., Fraire, J.A., Garello, R., 2022. Sparse satellite constellation design for global and regional direct-to-satellite IoT services. IEEE Transactions on Aerospace and Electronic Systems 58, 3786–3801. doi:10.1109/taes.2022.3185970.

Centenaro, M., Costa, C.E., Granelli, F., Sacchi, C., Vangelista, L., 2021. A survey on technologies, standards and open challenges in satellite IoT. IEEE Communications Surveys & Tutorials 23, 1693–1720. doi:10.1109/comst.2021.3078433.

Chen, X., Liao, X., Bidokhti, S.S., 2021. Real-time sampling and estimation on random access channels: Age of information and beyond, in: IEEE INFOCOM 2021 - IEEE Conference on Computer Communications, IEEE. pp. 1–10. doi:10.1109/infocom42981.2021.9488702.

Cheng, R., Kalashnikov, D.V., Prabhakar, S., 2003. Evaluating probabilistic queries over imprecise data, in: Proceedings of the ACM SIGMOD International Conference on Management of Data.

Cheng, R., Prabhakar, S., 2003. Managing uncertainty in sensor database. ACM SIGMOD Record 32, 41–46. doi:10.1145/959060.959068.

Choi, Y.D., Goodall, J.L., Sadler, J.M., Castronova, A.M., Bennett, A., Li, Z., Nijssen, B., Wang, S., Clark, M.P., Ames, D.P., Horsburgh, J.S., Yi, H., Bandaragoda, C., Seul, M., Hooper, R., Tarboton, D.G., 2021. Toward open and reproducible environmental modeling by integrating online data repositories, computational environments, and model application programming interfaces. Environmental Modelling & Software 135, 104888. doi:10.1016/j.envsoft.2020.104888.

Deligiannakis, A., Kotidis, Y., Roussopoulos, N., 2004. Hierarchical in-network data aggregation with quality guarantees, in: Advances in Database Technology - EDBT 2004. Springer Berlin Heidelberg, Berlin, Heidelberg. volume 2992 of *Lecture Notes in Computer Science*, pp. 658–675. doi:10.1007/978-3-540-24741-8_38.

Deligiannakis, A., Kotidis, Y., Roussopoulos, N., 2007. Dissemination of compressed historical information in sensor networks. The VLDB Journal 16, 439–461. doi:10.1007/s00778-005-0173-5.

Deshpande, A., Guestrin, C., Madden, S., Hellerstein, J.M., Hong, W., 2004. Model-driven data acquisition in sensor networks, in: Proceedings of the 30th International Conference on Very Large Data Bases (VLDB 2004), pp. 588–599. doi:10.1016/B978-012088469-8.50053-X.

Diaz-Cacho, M., Delgado, E., Barreiro, A., Falcón, P., 2017. Basic send-on-delta sampling for signal tracking-error reduction. Sensors 17, 312.

Duncan, A.E., 2020. The dangerous couple: Illegal mining and water pollution—a case study in Fena river in the Ashanti region of Ghana. Journal of Chemistry 2020, 1–9. doi:10.1155/2020/2378560.

Elliott, E.O., 1963. Estimates of error rates for codes on burst-noise channels. Bell System Technical Journal 42, 1977–1997. doi:10.1002/j.1538-7305.1963.tb00955.x.

Elmeleegy, H., Elmagarmid, A.K., Cecchet, E., Aref, W.G., Zwaenepoel, W., 2009. Online piece-wise linear approximation of numerical streams with precision guarantees. Proceedings of the VLDB Endowment 2, 145–156. doi:10.14778/1687627.1687645.

Essawy, B.T., Goodall, J.L., Voce, D., Morsy, M.M., Sadler, J.M., Choi, Y.D., Tarboton, D.G., Malik, T., 2020. A taxonomy for reproducible and replicable

research in environmental modelling. Environmental Modelling & Software 134, 104753. doi:10.1016/j.envsoft.2020.104753.

Gilbert, E.N., 1960. Capacity of a burst-noise channel. Bell System Technical Journal 39, 1253–1265. doi:10.1002/j.1538-7305.1960.tb03959.x.

Horsburgh, J.S., Reeder, S.L., Jones, A.S., Meline, J., 2015. Open source software for visualization and quality control of continuous hydrologic and water quality sensor data. Environmental Modelling & Software 70, 32–44. doi:10.1016/j.envsoft.2015.04.002.

Hutton, C., Wagener, T., Freer, J., Han, D., Duffy, C., Arheimer, B., 2016. Most computational hydrology is not reproducible, so is it really science?: REPRODUCIBLE COMPUTATIONAL HYDROLOGY. Water Resources Research 52, 7548–7555. doi:10.1002/2016wr019285.

Imai, H., Iri, M., 1988. Polygonal approximations of a curve: formulations and algorithms, in: Toussaint, G.T. (Ed.), Computational Morphology. North-Holland, Amsterdam, pp. 71–86.

Imielinski, T., Lipski, W., 1984. Incomplete information in relational databases. Journal of the ACM 31, 761–791. doi:10.1145/1634.1886.

Jakeman, A., Letcher, R., Norton, J., 2006. Ten iterative steps in development and evaluation of environmental models. Environmental Modelling & Software 21, 602–614. doi:10.1016/j.envsoft.2006.01.004.

Kamra, A., Misra, V., Feldman, J., Rubenstein, D., 2006. Growth codes: maximizing sensor network data persistence. ACM SIGCOMM Computer Communication Review 36, 255–266. doi:10.1145/1151659.1159943.

Kinéis, 2023. KIM1 Kinéis module: product datasheet (KINEIS-MU-22-0032). Kinéis, Ramonville-Saint-Agne, France. Accessed 23 October 2026.

Kinéis, 2024. Distribution formats: JSON formats (help centre). https://kineis.helpjuice.com/distribution-formats/json-formats. Accessed 25 October 2026.

Lin, C., Boursier, E., Papakonstantinou, Y., 2019a. Plato: Approximate analytics over compressed time series with tight deterministic error guarantees. ArXiv preprint arXiv:1808.04876.

Lin, J.W., Liao, S.w., Leu, F.Y., 2019b. Sensor data compression using bounded error piecewise linear approximation with resolution reduction. Energies 12, 2523. doi:10.3390/en12132523.

Lin, J.W., Liao, S.w., Tsai, Y.H., Huang, C.C., 2024. Progressive bounded error piecewise linear approximation with resolution reduction for time series data compression. Sensors 25, 145. doi:10.3390/s25010145.

Muñiz-Cuza, C.E., Boehm, M., Pedersen, T.B., 2025. CAMEO: Autocorrelation-preserving line simplification for lossy time series compression. ArXiv preprint arXiv:2501.14432.

Olston, C., Jiang, J., Widom, J., 2003. Adaptive filters for continuous queries over distributed data streams, in: Proceedings of the 2003 ACM SIGMOD international conference on Management of data, ACM. pp. 563–574. doi:10.1145/872757.872825.

O'Rourke, J., 1981. An on-line algorithm for fitting straight lines between data ranges. Communications of the ACM 24, 574–578. doi:10.1145/358746.358758.

Photopoulou, T., Fedak, M.A., Matthiopoulos, J., McConnell, B., Lovell, P., 2015. The generalized data management and collection protocol for conductivity-temperature-depth satellite relay data loggers. Animal Biotelemetry 3, 21. doi:10.1186/s40317-015-0053-8.

Rathi, S., Borkotoky, S.S., 2025. Message replication for improving reliability of LR-FHSS direct-to-satellite IoT. doi:10.48550/ARXIV.2501.11984. presented at the IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS 2024); arXiv preprint arXiv:2501.11984.

Riva, G.G., Finochietto, J.M., 2014. In-network filtering schemes for type-threshold function computation in wireless sensor networks. International Journal of Distributed Sensor Networks 10, 245924. doi:10.1155/2014/245924.

Rode, M., Wade, A.J., Cohen, M.J., Hensley, R.T., Bowes, M.J., Kirchner, J.W., Arhonditsis, G.B., Jordan, P., Kronvang, B., Halliday, S.J., Skeffington, R.A., Rozemeijer, J.C., Aubert, A.H., Rinke, K., Jomaa, S., 2016. Sensors in the stream: The high-frequency wave of the present. Environmental Science & Technology 50, 10297–10307. doi:10.1021/acs.est.6b02155.

Silberstein, A., Braynard, R., Yang, J., 2006. Constraint chaining: on energy-efficient continuous monitoring in sensor networks, in: Proceedings of the 2006 ACM SIGMOD International Conference on Management of Data, ACM. pp. 157–168. doi:10.1145/1142473.1142492.

Silberstein, A., Puggioni, G., Gelfand, A., Munagala, K., Yang, J., 2007. Suppression and failures in sensor networks: a Bayesian approach, in: Proceedings of the 33rd International Conference on Very Large Data Bases (VLDB 2007), pp. 842–853.

Soroush, E., Wu, K., Pei, J., 2008. Fast and quality-guaranteed data streaming in resource-constrained sensor networks, in: Proceedings of the 9th ACM international symposium on Mobile ad hoc networking and computing, ACM. pp. 391–400. doi:10.1145/1374618.1374670.

Sun, G., Karras, P., Zhang, Q., 2025. Highly efficient direct analytics on semantic-aware time series data compression. ArXiv preprint arXiv:2503.13246.

Talli, P., Munari, A., Mason, F., Chiariotti, F., Zanella, A., 2026. Value-based massive access through goal-oriented irregular repetition slotted ALOHA. doi:10.48550/ARXIV.2609.20569. arXiv preprint arXiv:2609.20569.

Taurone, F., Feh'er, M., Sipos, M., Lucani, D.E., 2024. TREAT: Two wrongs make a right: efficient distributed storage and queries of IoT datasets with erasure coding and compression, in: Proceedings of the 18th ACM International Conference on Distributed and Event-based Systems (DEBS). doi:10.1145/3629104.3666039.

Tondo, F.A., Montejo-Sánchez, S., Pellenz, M.E., Céspedes, S., Souza, R.D., 2021. Direct-to-satellite IoT slotted Aloha systems with multiple satellites and unequal erasure probabilities. Sensors 21, 7099. doi:10.3390/s21217099.

Tulone, D., Madden, S., 2006. PAQ: time series forecasting for approximate query answering in sensor networks, in: Wireless Sensor Networks (EWSN 2006). Springer. volume 3868 of *Lecture Notes in Computer Science*, pp. 21–37. doi:10.1007/11669463_5.

U.S. Environmental Protection Agency, 1986a. Ambient water quality criteria for dissolved oxygen. Technical Report EPA 440/5-86-003. Office of Water Regulations and Standards. Washington, DC, USA.

U.S. Environmental Protection Agency, 1986b. Quality criteria for water 1986. Technical Report EPA 440/5-86-001. Office of Water Regulations and Standards. Washington, DC, USA.

U.S. Environmental Protection Agency, 2003. EPA Region 10 guidance for Pacific Northwest state and tribal temperature water quality standards. Technical Report EPA 910-B-03-002. Region 10 Office of Water. Seattle, WA, USA.

U.S. Geological Survey, 2016. National water information system data available on the world wide web (USGS water data for the nation). doi:10.5066/F7P55KJN. instantaneous values for 2023 and 2024, accessed October 2026.

Wagner, R.J., Boulger, Jr., R.W., Oblinger, C.J., Smith, B.A., 2006. Guidelines and standard procedures for continuous water-quality monitors: station operation, record computation, and data reporting. Techniques and Methods 1-D3. U.S. Geological Survey. Reston, VA, USA. doi:10.3133/tm1D3.

Yates, R.D., Sun, Y., Brown, D.R., Kaul, S.K., Modiano, E., Ulukus, S., 2021. Age of information: An introduction and survey. IEEE Journal on Selected Areas in Communications 39, 1183–1210. doi:10.1109/jsac.2021.3065072.

Zhang, P., Liu, Y., Song, Y., Zhang, J., 2023. Advances and challenges in semantic communications: a systematic review. National Science Open , 20230029doi:10.1360/nso/20230029.